
SFT Memorizes, RL Generalizes: A Comparative Study of Foundation Model Post-training

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Abstract

Supervised fine-tuning (SFT) and reinforcement learning (RL) are widely used post-training techniques for foundation models. However, their respective role in enhancing model generalization in rule-based reasoning tasks remains unclear. This paper studies the comparative effect of SFT and RL on generalization and memorization, focusing on text-based and visual reasoning tasks. We introduce *GeneralPoints*, an arithmetic reasoning card game, and also consider *V-IRL*, a real-world navigation environment, to assess how models trained with SFT and RL generalize to unseen variants in both novel textual rules and visual domains. We show that RL, especially when trained with an outcome-based reward, generalizes in both the rule-based textual and visual environments. SFT, in contrast, tends to memorize the training data and struggles to generalize out-of-distribution in either scenario. Further analysis reveals that RL improves the model’s underlying visual recognition capabilities, contributing to its enhanced generalization in visual domains. Despite RL’s superior generalization, we show that SFT is still helpful for effective RL training: SFT stabilizes the model’s output format, enabling subsequent RL to achieve its performance gains. These findings demonstrate the advantage of RL for acquiring generalizable knowledge in complex, multi-modal tasks.

1. Introduction

Although SFT and RL are both widely used for foundation model training (OpenAI, 2023b; Google, 2023; Jaech et al., 2024; DeepSeekAI et al., 2025), their distinct effects on *generalization* (Bousquet & Elisseeff, 2000; Zhang et al., 2021) remain unclear, making it challenging to build reliable and robust AI systems. A key challenge in analyzing the generalizability of foundation models (Bommasani et al., 2021; Brown et al., 2020) is to separate data memorization¹ from the acquisition of transferable principles. Thus, we investigate the key question whether SFT or RL primarily memorize training data (Allen-Zhu & Li, 2023a; Ye et al., 2024; Kang et al., 2024), or whether they learn generalizable rules that can adapt to novel task variants.

To address this question, we focus on two aspects of generalization: textual rule-based generalization and visual generalization. For textual rules, we study the ability of a model to apply learned rules (given text instructions) to variants of these rules (Zhu et al., 2023; Yao et al., 2024; Ye et al., 2024). For vision-language models (VLMs), visual generalization measures the consistency of performance with variations in visual input, such as color and spatial layout, within a given task. For studying text-based and visual generalization, we investigate two different tasks that embody rule-based and visual variants. Our first task is *GeneralPoints*, an original card game task similar to *Points24* of RL4VLM (Zhai et al., 2024a), which is designed to evaluate a model’s *arithmetic reasoning capabilities*. The model receives four cards (presented as a text description or an image), and is required to compute a target number (24 by default) using each card’s numerical value exactly once. Second, we adopt *V-IRL* (Yang et al., 2024a), a real-world navigation task that focuses on the model’s *spatial reasoning capabilities*.

We adopt a multi-step RL framework similar to Zhai et al. (2024a), by instantiating RL after running SFT on

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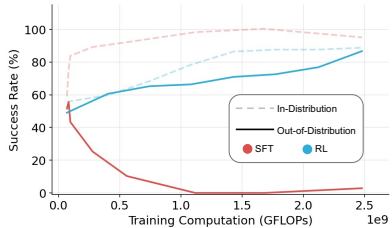


Figure 1: **A comparative study of RL and SFT on the visual navigation environment V-IRL (Yang et al., 2024a) for OOD generalization.** OOD curves represent performance on the same task, using a *different textual action space*. See detailed descriptions of the task in Section 5.1.

the backbone model (Dubey et al., 2024), using the sequential revision formulation (Snell et al., 2024). In both GeneralPoints and V-IRL, we observe that RL learns generalizable rules (expressed in text), where in-distribution performance gains also transfer to unseen rules. In contrast, SFT appears to memorize the training rules and does not generalize (see Figure 1 for an example). Beyond textual rule-based generalization, we further investigate generalization in the visual domain and observe that RL also generalizes to visual OOD tasks, whereas SFT continues to struggle. As a by-product of the visual OOD generalization capability, our multi-turn RL approach **achieves state-of-the-art performance** on the V-IRL mini benchmark, by **+33.8%** (44.0%→77.8%) (Yang et al., 2024a), highlighting the generalization capability of RL. To understand how RL affects the visual abilities of a model, we conducted additional analysis on GeneralPoints, revealing that training RL with an outcome-based reward function (Cobbe et al., 2021) improves visual recognition capabilities. Although RL exhibits superior generalization compared to SFT, we show that SFT is still necessary to stabilize the model’s output format, enabling RL to achieve its performance gains. Last but not least, we observe that scaling up the inference time compute by increasing the number of maximal steps leads to better generalization.

2. Related Works

Post-training. Post-training is crucial for enhancing model performance (Zhang et al., 2022; Hoffmann et al., 2023; OpenAI, 2023b; Google, 2023; Touvron et al., 2023). This stage commonly utilizes large-scale supervised fine-tuning (SFT) (Radford et al., 2018; Brown et al., 2020; Radford et al., 2021; Wei et al., 2022a; Chung et al., 2022; Zhou et al., 2024a) and/or reinforcement learning (RL) (Ziegler et al., 2019; Ouyang et al., 2022; Sun et al., 2024; Abdulhai et al., 2023; Zhou et al., 2024b; Zhai et al.,

2024a). SFT adapts pre-trained models to downstream tasks by training them on task-specific, often instruction-formatted datasets. Previous work, such as FLAN (Wei et al., 2022a), demonstrates that fine-tuning on diverse instruction-tuning datasets significantly enhances zero-shot performance on unseen tasks. Furthermore, LIMA (Zhou et al., 2024a) shows that supervised fine-tuning acts as a “format teacher” effectively adapting the model’s responses to a desired format while leveraging the capabilities of pre-trained LLMs. In contrast, RL (Ziegler et al., 2019; Ouyang et al., 2022; Sun et al., 2024; Ramamurthy et al., 2023; Abdulhai et al., 2023; Zhou et al., 2024b; Zhai et al., 2024a) has been primarily used to align models with human preferences or training the foundational model to solve a specific task (Abdulhai et al., 2023; Zhou et al., 2024b; Zhai et al., 2024a; Chen et al., 2024b). Our work differs from prior studies, as we aim to comparatively analyze the generalization and memorization of SFT and RL on *both LLM and VLM*, while previous studies have focused primarily on only one of these two post-training methods (or only study LLM or VLM) or on only one post-training method.

Memorization and generalization in LLM/VLM. Several studies have examined the interplay between memorization and generalization in neural networks (Han et al., 2022; Carlini et al., 2022; Yang et al., 2023). In LLMs, memorization can manifest as the model memorizing the training data (Carlini et al., 2022; Jiang et al., 2024; Kang et al., 2024), while generalization reflects the divergence between the model’s output distribution and the pre-training data distribution (Zhang et al., 2023). Prior studies suggest that LLMs exhibit more overfitting on simpler, knowledge-intensive tasks and greater generalization on more complex, reasoning-intensive ones (Wang et al., 2024; Qi et al., 2024). For example, recent studies (Ye et al., 2024; Allen-Zhu, 2024; Allen-Zhu & Li, 2023a; Tong et al., 2024b) have demonstrated that LLMs develop reasoning skill sets beyond their training data by pre-computing reasoning graphs before autoregressive generation, which provides compelling evidence of generalization. Our study takes a different approach by investigating the role of different post-training paradigms on memorization versus generalization in the context of textual rule-based and visual variants. We conduct comparative studies in both unimodal (LLM) and multimodal (VLM) settings, and demonstrate that RL leads to better generalization performance than SFT.

Scaling up inference-time compute. Recent research has increasingly focused on scaling up inference-time computation to improve model performance (Wei et al., 2022b; Yao et al., 2024; Snell et al., 2024; Jaech et al., 2024). Early studies (Wei et al., 2022b; Yao et al., 2024) prompted

SFT 记忆, RL 泛化: 基础模型后训练的对比研究

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Abstract

化知识方面的优势。

1. Introduction

尽管监督微调 (SFT) 和强化学习 (RL) 在基础模型训练中均被广泛使用 (OpenAI, 2023b; Google, 2023; Jaech et al., 2024; DeepSeekAI et al., 2025), 但它们对泛化 (Bousquet & Elisseff, 2000; Zhang et al., 2021) 的独特影响仍不明确, 这使得构建可靠且稳健的人工智能系统面临挑战。分析基础模型 (Bommasani et al., 2021; Brown et al., 2020) 泛化能力的一个关键挑战在于区分数据记忆¹与可迁移原理的习得。因此, 我们探究一个核心问题: SFT 或 RL 主要是记忆训练数据 (Allen-Zhu & Li, 2023a; Ye et al., 2024; Kang et al., 2024), 还是学习能够适应新任务变体的可泛化规则。

为解答此问题, 我们聚焦于泛化的两个方面: 基于文本规则的泛化和视觉泛化。对于文本规则, 我们研究模型将习得的规则 (通过文本指令给出) 应用于这些规则变体 (Zhu et al., 2023; Yao et al., 2024; Ye et al., 2024) 的能力。对于视觉语言模型 (VLMs), 视觉泛化衡量的是在给定任务中, 模型性能随视觉输入 (如颜色和空间布局) 变化的一致性。为研究基于文本的泛化和视觉泛化, 我们考察了体现规则变体和视觉变体的两项不同任务。我们的第一项任务是 GeneralPoints, 这是一种类似于 RL4VLM 中 Points24 (Zhai et al., 2024a) 的原创卡牌游戏任

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models to generate intermediate reasoning steps and extend the responses before producing a final answer. Subsequent work (Zelikman et al., 2022; Feng et al., 2023; Tian et al., 2024; Chen et al., 2024a; Snell et al., 2024) has demonstrated that fine-tuning verifiers during inference improves model accuracy, effectively utilizing test-time computation. Notably, recent findings (Jaech et al., 2024; DeepSeekAI et al., 2025) reveal “scaling laws” for inference-time compute, highlighting significant performance gains with increased computational resources. Our work builds upon these findings in two ways. First, we integrate insights from inference-time verification into a multi-turn RL formulation that allows the model to identify and correct its errors. Second, we examine the impact of inference-time verification on RL generalization, demonstrating that scaling up inference-time verification (in terms of the maximum number of verification steps) is a key for RL to generalize.

Improving visual capability in VLMs. While VLMs have demonstrated remarkable skill across a wide range of challenging tasks, such as solving advanced college exam questions (Lu et al., 2023; Yue et al., 2024a;b) and spatial understanding tasks (Yang et al., 2024a;b), they also exhibit limitations in visual perception (Zhai et al., 2024a;b; Tong et al., 2024c;d; Rahmanzadehgervi et al., 2024). Prior efforts to enhance VLMs’ visual perception include combining multiple visual encoders (Tong et al., 2024d; Kar et al., 2025; Tong et al., 2024a), curating high-quality SFT data (Chen et al., 2023; Liu et al., 2024; Tong et al., 2024a), and improving the SFT training recipe by unfreezing the visual backbone (Liu et al., 2023; Tong et al., 2024a). While these prior works primarily focus on experiments during the SFT stage, our work demonstrates that RL can also improve visual perception.

3. Preliminaries

Standard RL terminology. We consider finite horizon decision making, and adopt standard notation from the classical RL literature (Sutton & Barto, 2018; Agarwal et al., 2019), where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ denotes the action space, $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ denotes the reward function, and T denotes the maximum number of steps per episode. The goal is to learn a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ that maximizes the overall return $\max_{\pi \in \Pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T r_t \right]$, where r_t denotes $r(s_t, a_t)$. Without loss of generality, we use $\pi(a|s) \in [0, 1]$ to denote probability of π choosing a at s .

Adapting RL terminology to LLM/VLM with a verifier. We adopt a multi-turn RL setting for foundation model training (Zhai et al., 2024a). Let $\mathcal{V}$ represent the discrete and finite vocabulary (token) space. The input and output text spaces are denoted by $\mathcal{V}^m$ and $\mathcal{V}^n$ respectively, where

m and n are the maximum token length of the input sequence v^{in} and output sequence v^{out} . For models requiring visual inputs (VLM), we define $\mathcal{O}$ as the space of all RGB images. The state space, denoted by $\mathcal{S}$, is defined as $\mathcal{S} := \mathcal{V}^m \times \mathcal{O}$ for VLM, and $\mathcal{S} := \mathcal{V}^m$ for LLM. The action space $\mathcal{A}$ is defined as $\mathcal{A} := \mathcal{V}^n$. We use $\text{VER} : \mathcal{V}^n \rightarrow \mathbb{R} \times \mathcal{V}^k$ to denote a verifier, which evaluates the outcome of v^{out} and generates an outcome-based reward function (Cobbe et al., 2021; Hosseini et al., 2024; Snell et al., 2024; Setlur et al., 2024) r along with textual information v^{ver} . Mathematically, at time t , $\text{VER}(v_t^{\text{out}}) \mapsto (r_t, v_t^{\text{ver}})$. Similar to Zhai et al. (2024a), we treat the model with parameter θ as our policy network $\pi_{\theta} : \mathcal{S} \rightarrow \mathcal{V}^n$, and adopt PPO (Schulman et al., 2017) as the backbone RL algorithm for updating π_{θ} .

Sequential revision. For modeling the state-action transition, we adopt the sequential revision formulation (Snell et al., 2024). Specifically, at time step $t = 0$ the initial input v_0^{in} consists of the system prompt. For subsequent time steps ($t \geq 1$), the input prompt v_t^{in} comprises the system prompt concatenated with all prior model and verifier outputs, denoted by $[v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1}$. An illustration of the sequential revision is provided in Figure 2 (also see Figure 5 of Snell et al. (2024)), and an example of the state-action transition is shown in Figure 3.

4. Evaluation Tasks

To evaluate the generalization of different post-training methods, we select two tasks that each offer *rule* and *visual variations*. The first task, *GeneralPoints*, is a new environment we have designed that allows assessment of arithmetic reasoning abilities (Section 4.1). The second task, *V-IRL* (Yang et al., 2024a), is chosen to examine the model’s reasoning capabilities in an open-world visual navigation domain (Section 4.2).

4.1. The General Points Environment

Our original *GeneralPoints* environment, instantiated on top of the *Points24* environment (Zhai et al., 2024a), is designed to evaluate generalization of arithmetic reasoning. Each state s of the environment contains 4 cards, described as text (in the *GP-T* variant) or presented as an image (in the *GP-VL* variant); see Figure 2 left for a visual example of *GeneralPoints*. The goal is to *produce an equation that equals a target number* (24 by default) using all 4 numbers from the cards *exactly once*. Detailed examples of the state-action transitions are provided in Appendix A.2. Note that when input from *GeneralPoints* is presented in an image (*GP-VL*), it naturally introduces additional visual challenges requiring the VLM to *recognize all cards before solving the equation*.

务，旨在评估模型的算术推理能力。模型接收四张卡牌（以文本描述或图像形式呈现），并需使用每张卡牌的数字值恰好一次来计算目标数字（默认为24）。其次，我们采用 *V-IRL* (Yang et al., 2024a)，这是一项关注模型空间推理能力的真实世界导航任务。我们采用与 Zhai et al. (2024a) 类似的多步强化

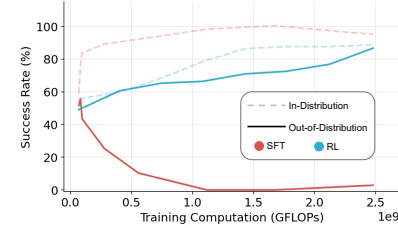


Figure 1: 在视觉导航环境 *V-IRL* (Yang et al., 2024a) 上对 *RL* 与 *SFT* 进行 OOD 泛化的比较研究。OOD 曲线表示使用不同的文本动作空间在同一任务上的性能表现。任务详细描述见第 5 节。学习框架，通过在骨干模型 (Dubey et al., 2024) 上进行监督微调后实例化强化学习，并采用顺序修订的表述方式 (Snell et al., 2024)。在 *GeneralPoints* 和 *V-IRL* 中，我们观察到强化学习能够学习可泛化的规则（以文本形式表达），其分布内性能增益也能迁移到未见规则上。相比之下，监督微调似乎只是记住了训练规则，并未展现出泛化能力（示例参见 Figure 1）。除了基于文本规则的泛化，我们进一步探究了视觉领域的泛化表现，发现强化学习同样能泛化至视觉分布外任务，而监督微调则持续面临困难。作为视觉分布外泛化能力的副产品，我们的多轮强化学习方法在 *V-IRL* 迷你基准测试中取得了最先进的性能，提升幅度达 +33.8% (44.0% → 77.8%) (Yang et al., 2024a)，这凸显了强化学习的泛化能力。为理解强化学习如何影响模型的视觉能力，我们对 *GeneralPoints* 进行了补充分析，发现采用基于结果的奖励函数 (Cobbe et al., 2021) 进行强化学习训练能提升视觉识别能力。尽管强化学习相比监督微调展现出更优的泛化性能，但我们证明监督微调仍是必要的，它能稳定模型的输出格式，从而使强化学习得以实现性能增益。最后同样重要的是，我

们观察到通过增加最大步数来扩展推理时的计算量，能够带来更好的泛化效果。

2. Related Works

Post-training. 后训练对于提升模型性能至关重要 (Zhang et al., 2022; Hoffmann et al., 2023; OpenAI, 2023b; Google, 2023; Touvron et al., 2023)。这一阶段通常利用大规模有监督微调 (SFT) (Radford et al., 2018; Brown et al., 2020; Radford et al., 2021; Wei et al., 2022a; Chung et al., 2022; Zhou et al., 2024a) 和/或强化学习 (RL) (Ziegler et al., 2019; Ouyang et al., 2022; Sun et al., 2024; Abdulhai et al., 2023; Zhou et al., 2024b; Zhai et al., 2024a)。SFT 通过在任务特定、通常是指令格式的数据集上进行训练，使预训练模型适应下游任务。先前的工作，如 FLAN (Wei et al., 2022a)，证明了在多样化的指令调优数据集上进行微调能显著提升模型在未见任务上的零样本性能。此外，LIMA (Zhou et al., 2024a) 表明，有监督微调充当了一种“格式教师”，有效地将模型的响应调整到期望的格式，同时利用了预训练大语言模型的能力。相比之下，RL (Ziegler et al., 2019; Ouyang et al., 2022; Sun et al., 2024; Ramamurthy et al., 2023; Abdulhai et al., 2023; Zhou et al., 2024b; Zhai et al., 2024a) 主要用于使模型与人类偏好对齐，或训练基础模型以解决特定任务 (Abdulhai et al., 2023; Zhou et al., 2024b; Zhai et al., 2024a; Chen et al., 2024b)。我们的工作与先前研究不同，因为我们旨在比较分析 SFT 和 RL 在大语言模型和视觉语言模型上的泛化与记忆能力，而先前研究主要只关注这两种后训练方法之一（或仅研究大语言模型或视觉语言模型），或仅关注一种后训练方法。

Memorization and generalization in LLM/VLM. 多项研究探讨了神经网络中记忆与泛化之间的相互作用 (Han et al., 2022; Carlini et al., 2022; Yang et al., 2023)。在大语言模型中，记忆可能表现为模型记住了训练数据 (Carlini et al., 2022; Jiang et al., 2024; Kang et al., 2024)，而泛化则反映了模型输出分布与预训练数据分布之间的差异 (Zhang et al., 2023)。先

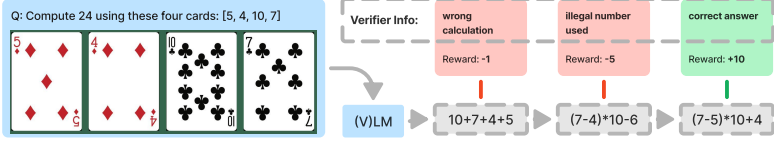


Figure 2: An example of the sequential revision formulation with a verifier. The model generate the next answer v_{t+1}^{out} conditioned on all previous answers and information ($v_i^{\text{out}}, v_i^{\text{ver}}, 0 \leq i \leq t$) from the verifier.

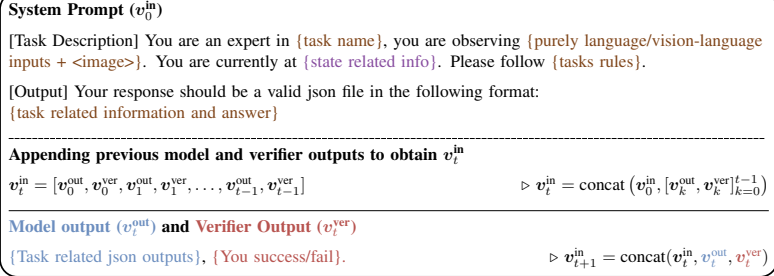


Figure 3: A template of our prompt update for constructing v_{t+1}^{in} . The brown parts marks the task and related information, and the purple parts denote the state (s_t) specific info. The blue and red describe the output from the model and verifier, respectively.

Rule variations. To study whether the model learns arithmetic operations or simply memorizes the post-training data, we introduce rule variations in GeneralPoints. These variations consist of interpreting the symbols 'J', 'Q', and 'K' either as '11', '12', and '13', respectively, or all as the same number '10'. These variations ensure a rigorous evaluation of the model's ability to generalize arithmetic reasoning across diverse settings. Each rule is specified as text in the input prompt, see the {tasks rules} part in Figure 3. For studying ruled based generalization, we post-train the model using one rule, then evaluate using a different rule.

Visual variations. The GeneralPoints environment can also be naturally customized to evaluate generalization across visual variants. Since the major visual challenge is to recognize the number of each card, agnostic to the color of the cards, we consider the cards with different colors as visual variants of the task. In the visual generalization setting, we train the model using cards of one color, then test OOD performance using the other color.

4.2. The V-IRL Environment

While the GeneralPoints environment is designed to assess arithmetic reasoning abilities, we further utilize the V-IRL environment (Yang et al., 2024a) to study spatial reasoning ability in an open-world navigation domain that

uses realistic visual input. As in GeneralPoints we consider two versions of the environment, one (V-IRL-L) that consists of pure language descriptions,² and another (V-IRL-VL) that includes vision-language input. The major visual challenge in V-IRL involves recognizing different landmarks from the visual observation³ before taking an action. The goal is to navigate to a target location by following a set of instructions that contain spatial information. A detailed example of one environment step is shown in Appendix B.2.

Rule variations. To evaluate whether the model possesses spatial knowledge or simply memorizes post-training data, we consider two distinct action space configurations. The first variant utilizes an absolute orientation action space, which includes {'north', 'northeast', 'east', 'southeast', 'south', 'southwest', 'west', 'northwest'}. The second variant employs a relative orientation action space, containing {'left', 'right', 'slightly left', 'slightly right'}. This relative configuration adjusts the current orientation by 90 degrees or 45 degrees to the left or right, respectively. An

²The visual input can be parsed into pure text description, see more details in Yang et al. (2024a) and an illustration of pure text the version in Figure 14.

³See Figure 4, the model needs to recognize landmarks like The Dutch, Lola Taverna, and Shuka from the visual observation, and relate these landmarks with the textual instructions for taking the right action.

前研究表明, 大语言模型在更简单、知识密集型的任务上表现出更多的过拟合, 而在更复杂、推理密集型的任务上表现出更强的泛化能力 (Wang et al., 2024; Qi et al., 2024). 例如, 最近的研究 (Ye et al., 2024; Allen-Zhu, 2024; Allen-Zhu & Li, 2023a;b; 2024; Tong et al., 2024b) 表明, 大语言模型通过在自回归生成前预计算推理图, 发展出了超越其训练数据的推理技能集, 这为泛化提供了有力的证据。我们的研究采取了不同的方法, 通过调查不同后训练范式的基于文本规则和视觉变体背景下对记忆与泛化的作用。我们在单模态 (大语言模型) 和多模态 (视觉语言模型) 设置下进行了比较研究, 并证明强化学习比有监督微调带来更好的泛化性能。

Scaling up inference-time compute. 近期研究日益关注通过扩展推理时计算以提升模型性能 (Wei et al., 2022b; Yao et al., 2024; Snell et al., 2024; Jaech et al., 2024). 早期研究 (Wei et al., 2022b; Yao et al., 2024) 促使模型在生成最终答案前, 先生成中间推理步骤并扩展回应。后续工作 (Zelikman et al., 2022; Feng et al., 2023; Tian et al., 2024; Chen et al., 2024a; Snell et al., 2024) 表明, 在推理过程中微调验证器能有效利用测试时计算, 从而提高模型准确性。值得注意的是, 最新发现 (Jaech et al., 2024; DeepSeekAI et al., 2025) 揭示了推理时计算的“缩放定律”, 突显了计算资源增加带来的显著性能提升。我们的研究从两个方面基于这些发现展开。首先, 我们将推理时验证的见解整合到多轮强化学习框架中, 使模型能够识别并纠正自身错误。其次, 我们考察了推理时验证对强化学习泛化能力的影响, 证明扩展推理时验证 (就最大验证步数而言) 是强化学习实现泛化的关键。

Improving visual capability in VLMs. 尽管视觉语言模型已在众多挑战性任务中展现出卓越能力, 例如解答高等大学考试试题 (Lu et al., 2023; Yue et al., 2024a;b) 和空间理解任务 (Yang et al., 2024a;b), 但它们在视觉感知方面仍存在局限 (Zhai et al., 2024a;b; Tong et al., 2024c;d; Rahmanzadehgervi et al., 2024). 先前提视觉语言模型视觉感知的努力包括融合多

个视觉编码器 (Tong et al., 2024d; Kar et al., 2025; Tong et al., 2024a)、筛选高质量监督微调数据 (Chen et al., 2023; Liu et al., 2024; Tong et al., 2024a), 以及通过解冻视觉主干网络改进监督微调训练方案 (Liu et al., 2023; Tong et al., 2024a)。虽然这些先前工作主要聚焦于监督微调阶段的实验, 但我们的研究表明强化学习同样能改善视觉感知。

3. Preliminaries

Standard RL terminology. 我们考虑有限域决策问题, 并采用经典强化学习文献 (Sutton & Barto, 2018; Agarwal et al., 2019) 中的标准符号, 其中 $\mathcal{S}$ 表示状态空间, $\mathcal{A}$ 表示动作空间, $r: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ 表示奖励函数, T 表示每个回合的最大步数。目标是学习一个策略 $\pi: \mathcal{S} \rightarrow \mathcal{A}$, 以最大化总体回报 $\max_{\pi \in \Pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T r_t \right]$, 其中 r_t 表示 $r(s_t, a_t)$ 。不失一般性, 我们使用 $\pi(a|s) \in [0, 1]$ 表示在 s 时选择 a 的概率。

Adapting RL terminology to LLM/VLM with a verifier. 我们采用多轮强化学习设置进行基础模型训练 (Zhai et al., 2024a)。令 $\mathcal{V}$ 表示离散且有限的词汇 (词元) 空间。输入和输出文本空间分别由 $\mathcal{V}^m$ 和 $\mathcal{V}^n$ 表示, 其中 m 和 n 分别是输入序列 $\mathbf{v}^{\text{in}}$ 和输出序列 $\mathbf{v}^{\text{out}}$ 的最大词元长度。对于需要视觉输入的模式 (VLM), 我们定义 $\mathcal{O}$ 为所有 RGB 图像的空间。状态空间记为 $\mathcal{S}$, 对于 VLM 定义为 $\mathcal{S} := \mathcal{V}^m \times \mathcal{O}$, 对于 LLM 定义为 $\mathcal{S} := \mathcal{V}^m$ 。动作空间 $\mathcal{A}$ 定义为 $\mathcal{A} := \mathcal{V}^n$ 。我们使用 $\text{VER}: \mathcal{V}^n \rightarrow \mathbb{R} \times \mathcal{V}^k$ 表示一个验证器, 它评估 $\mathbf{v}^{\text{out}}$ 的结果并生成一个基于结果的奖励函数 (Cobbe et al., 2021; Hosseini et al., 2024; Snell et al., 2024; Setlur et al., 2024) r 以及文本信息 $\mathbf{v}^{\text{ver}}$ 。数学上, 在时间 t , $\text{VER}(\mathbf{v}_t^{\text{out}}) \mapsto (r_t, \mathbf{v}_t^{\text{ver}})$ 。类似于 Zhai et al. (2024a), 我们将参数为 θ 的模型视为我们的策略网络 $\pi_{\theta}: \mathcal{S} \rightarrow \mathcal{V}^n$, 并采用 PPO (Schulman et al., 2017) 作为更新 π_{θ} 的骨干强化学习算法。

Sequential revision. 为了对状态-动作转移进行建模, 我们采用顺序修订的表述 (Snell et al., 2024)。具体而言, 在时间步 $t = 0$, 初始输入 $\mathbf{v}_0^{\text{in}}$ 包含系统提



Figure 4: Demonstration of one navigation task in V-IRL. Agent navigates from place to place following the given linguistic navigation instructions in V-IRL. The navigation procedure is shown at the top, with the navigation instructions displayed below. Visual observation-related information is highlighted in green, while action-related information is marked in orange.

overview of a navigation task in V-IRL is provided in Figure 4, and a detailed state-action transition in V-IRL is provided in Figure 13 (in Appendix B.2).

Visual variations. The key visual challenge in V-IRL is to recognize landmarks from the visual observations (e.g., the green parts in Figure 4). Since the V-IRL environment contains visual observations from different cities, we can assess visual generalization in V-IRL by training the model to navigate in one location and then evaluate its performance in different locations.

5. Results

In this section, we present experiments that investigate the generalization abilities induced by post-training with RL and SFT. We adopt Llama-3.2-Vision-11B (Dubey et al., 2024) as the backbone model. Following the standard pipelines of RLHF (Ouyang et al., 2022) and RL4VLM (Zhai et al., 2024a), we initialize the model with SFT before running RL. We specifically study the follow-

ing questions. Section 5.1: how does SFT or RL affect the model’s generalization to different rules? Section 5.2: when the model contains a visual component, how does RL/SFT affect its generalization to different visual variants? Section 5.3: how does RL/SFT affect visual recognition capability in a VLM? Section 5.4: what role does SFT play in RL training? Section 5.5: how does the number of verification iterations affect generalization?

5.1. Generalization across Rules

We evaluate the performance of different post-training methods on GeneralPoints and V-IRL, each of which has a pure language (-L) and a vision-language (-VL) variant, and each encompassing rule variations. For each task, we separately scale the training compute for RL and SFT on a single rule. We consider the results on the trained rule as in-distribution (ID) performance, whereas results on the unseen rules measures out-of-distribution (OOD) generalization. In GeneralPoints, the ID case treats all ‘J’, ‘Q’, ‘K’ as 10, and the OOD case interprets them as 11, 12, and 13. As for V-IRL, the ID case adopts the *absolute orienta-*

tion. For subsequent time steps ($t \geq 1$), input prompts v_t^m by system prompts with all previous model and verifier outputs connected, denoted as $[v_k^m, v_k^v]_{k=0}^{t-1}$. Sequence editing diagrams are shown in Figure 2 (see Snell et al. (2024) Figure 5), state-action transition examples are shown in Figure 3.

4. Evaluation Tasks

For evaluation of the generalization ability of different training methods, we selected two tasks. Each task provides rules and visual variants. The first task GeneralPoints is our designed new environment for evaluating arithmetic reasoning ability (Section 4.1). The second task V-IRL (Yang et al., 2024a) aims to evaluate the model’s reasoning ability in the open-world visual navigation domain (Section 4.2).

4.1. The General Points Environment

We build on the Points24 environment (Zhai et al., 2024a) to construct the original GeneralPoints environment, designed for evaluating arithmetic reasoning generalization ability. Each state s contains 4 cards, represented by text (in GP-L variant) or images (in GP-VL variant); the visual examples of GeneralPoints can be seen in Figure 2. The goal is achieved by using all 4 cards’ numbers once, generating an equation equal to the target number (default 24). State-action transition details are in Appendix A.2. Note that when GeneralPoints’ input is presented in image form (GP-VL), it naturally introduces additional visual challenges, requiring VLM to solve equations before identifying all cards.

Rule variations. To explore whether the model is learning arithmetic or simply memorizing post-training data, we introduced rule variations in GeneralPoints. These variations include symbols ‘J’, ‘Q’, and ‘K’ being interpreted as ‘11’, ‘12’, and ‘13’, or all treated as the same number ‘10’. These variations strictly evaluate the model’s generalization ability across different settings. Each rule in the input prompt is specified in text form, see Figure 3 for the {task rules} part. For rule-based research, we use a rule-based model for training and evaluation, then use different rules for evaluation.

Visual variations. GeneralPoints environment can also be customized to evaluate cross-visual generalization. Since the main visual

challenge is to identify the numbers on each card, while the card color is irrelevant, we treat cards of different colors as visual variants. In the visual generalization setting, we use a color-based card training model, then use another color-based model to test its out-of-distribution performance.

4.2. The V-IRL Environment

Although GeneralPoints environment aims to evaluate arithmetic reasoning ability, we further utilize the V-IRL environment (Yang et al., 2024a) to study the model’s spatial reasoning ability in the open-world navigation domain. Similar to GeneralPoints, we consider two versions of the environment. One version (V-IRL-L) is composed of pure language descriptions, while another version (V-IRL-VL) includes visual-language input. V-IRL’s main visual challenge lies in identifying different landmarks from visual observations, then taking actions. The goal is to follow a set of instructions containing spatial information, navigate to the target location. Appendix B.2 shows a detailed step-by-step environment example.

Rule variations. To evaluate whether the model has spatial knowledge, we not only memorized the training data, but also considered two different action space configurations. The first variant uses absolute direction action space, including {‘N’, ‘NE’, ‘E’, ‘SE’, ‘S’, ‘SW’, ‘W’, ‘NW’}. The second variant uses relative direction action space, including {‘L’, ‘R’, ‘left’, ‘right’}. These relative configurations will adjust the forward direction left or right by 90 or 45 degrees. Figure 4 provides an overview of navigation tasks in V-IRL (see Appendix B.2) and provides a detailed state-action transition process.

Visual variations. V-IRL’s key visual challenge is to identify landmarks from visual observations (e.g., the green parts in Figure 4). Since the V-IRL environment contains visual observations from different cities, we can assess visual generalization in V-IRL by training the model to navigate in one location and then evaluate its performance in different locations.

²Visual input can be parsed into pure text descriptions, for more details please see Yang et al. (2024a), the text version diagram is shown in Figure 14.

³See Figure 4, the model needs to identify landmarks such as The Dutch, Lola Taverna and Shuka, and then associate these landmarks with the spatial information in the text instructions to take the correct action.

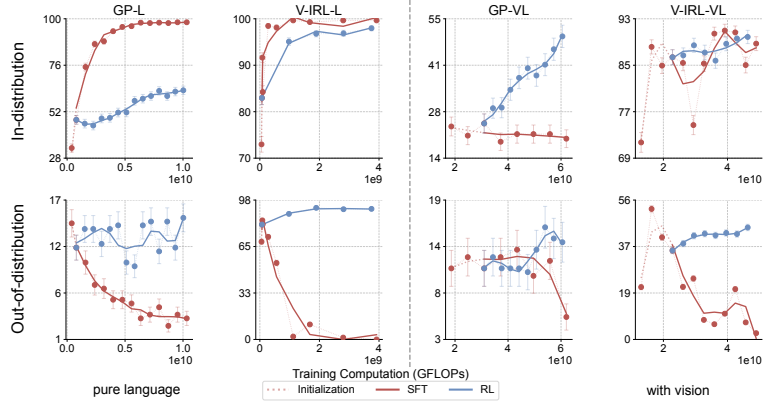


Figure 5: **Success rate (%) - GFLOPs trendlines for RL and SFT on GeneralPoints and V-IRL.** The top row shows in-distribution performance, while the bottom row shows out-of-distribution performance. Results are presented for both pure language (-L) and vision-language (-VL) variants of each task. For GeneralPoints, we report the episode success rate, while for V-IRL, we report per-step accuracy with overall success rate in Figures 1 and 18. Detailed evaluation setups (and curve smoothing) are provided in Appendix C.3.

tion coordinate system and the OOD case uses the *relative orientation* action space. Other details and additional experimental setup can be found in Appendix C.

RL generalizes, SFT memorizes. As illustrated in Figure 5, RL consistently improves OOD performance on all tasks, including both unimodal (LLM) and multi-modal (VLM). Specifically, Figure 6 demonstrates that RL achieves an increase of **+3.5%** on GP-L (11.5% $\rightarrow$ 15.0%) and **+11.0%** on V-IRL-L (80.8% $\rightarrow$ 91.8%). Even with the additional challenge of visual recognition in the VLM, RL maintains consistent performance improvements of **+3.0%** (11.2% $\rightarrow$ 14.2%) on GP-VL and **+9.3%** (35.7% $\rightarrow$ 45.0%) on V-IRL-VL, respectively. In contrast, SFT consistently exhibits performance degradation across all OOD evaluations on all tasks: **-8.1%** on GP-L (11.5% $\rightarrow$ 3.4%), **-79.5%** on V-IRL-L (80.8% $\rightarrow$ 1.3%), **-5.6%** (11.2% $\rightarrow$ 5.6%) on GP-VL, and **-33.2%** (35.7% $\rightarrow$ 2.5%) on V-IRL-VL.

5.2. Generalization in Visual Out-of-Distribution Tasks

Section 5.1 demonstrates that RL yields generalization across rule variations, whereas SFT exhibits the opposite trend. Since VLMs also incorporate a visual modality, we next study the effects of *visual* variation in OOD generalization. For GeneralPoints, we train the VLM using the black suits ($\clubsuit$, $\spadesuit$) and test out-of-distribution perfor-

mance on the red suits ($\heartsuit$, $\diamondsuit$). For V-IRL, we train the model on routes collected in New York City and evaluate it on the original V-IRL VLN mini benchmark (Yang et al., 2024a) containing routes from various cities worldwide (see Appendix B.1 for details). Note that the rules remain consistent across experiments in this section.

RL generalizes in visual OOD tasks. As shown in Figure 7, we observe that RL still *generalizes in visual OOD tasks*, while SFT continues to suffer. Specifically, in GP-VL and V-IRL-VL, RL achieves performance improvements of **+17.6%** (23.6% $\rightarrow$ 41.2%), **+61.1%** (16.7% $\rightarrow$ 77.8%), whereas SFT suffers from performance decreases of **-9.9%** (23.6% $\rightarrow$ 13.7%) and **-5.6%** (16.7% $\rightarrow$ 11.1%). As a byproduct of this visual OOD study, we also show that our multi-turn RL formulation **improves the state-of-the-art results** (see Table 5 of Yang et al. (2024a)) on the V-IRL mini benchmark by **+33.8%** (44.0% $\rightarrow$ 77.8%). Notably, unlike the previous state-of-the-art approach reported in V-IRL, which relies on a two stage VLM-LLM collaboration technique and tailored prompt engineering on closed-sourced model (OpenAI, 2023a), our end-to-end RL approach enables an open-sourced model (Dubey et al., 2024) to reach superior performance.

5.3. RL Improves Visual Capabilities

Building upon the above observation that VLMs trained with RL generalize to visual OOD tasks (Section 5.2), we

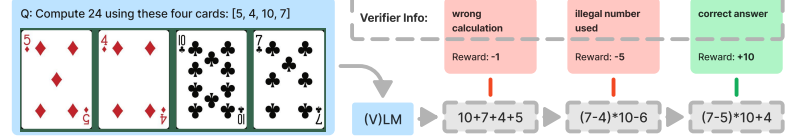


Figure 2: 一个带有验证器的顺序修订表述示例。模型基于验证器提供的所有先前答案和信息 ($v_t^{\text{out}}, v_t^{\text{ver}}, 0 \leq i \leq t$) 生成下一个答案 v_{t+1}^{out} 。

System Prompt (v_0^{in})

[任务描述] 您是{任务名称}方面的专家, 正在观察{纯语言/视觉-语言输入 + < 图像 >}。您当前处于{状态相关信息}。请遵循{任务规则}。[输出] 您的响应应是一个符合以下格式的有效 JSON 文件:

{任务相关信息与答案}

Appending previous model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \quad \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=1}^{t-1})$$

Model output (v_t^{out}) 和 验证器输出 (v_t^{ver})

$$\{\text{任务相关的 JSON 输出}\}, \{\text{您成功/失败}\}. \quad \triangleright v_{t+1}^{\text{in}} = \text{concat}(v_t^{\text{in}}, [v_t^{\text{out}}, v_t^{\text{ver}}])$$

Figure 3: 我们用于构建 v_{t+1}^{in} 的提示更新模板。其中棕色部分标记任务及相关信息, 紫色部分表示状态 (s_t) 特定信息。蓝色和红色分别描述来自模型和验证器的输出。

5. Results

在本节中, 我们展示研究 RL 和 SFT 后训练所引发的泛化能力的实验。我们采用 Llama-3.2-Vision-11B (Dubey et al., 2024) 作为骨干模型。遵循 RLHF (Ouyang et al., 2022) 和 RL4VLM (Zhai et al., 2024a) 的标准流程, 我们在运行 RL 之前先用 SFT 初始化模型。我们具体研究以下问题。Section 5.1: 当模型包含视觉组件时, RL/SFT 如何影响其针对不同视觉变量的泛化能力? Section 5.2: RL/SFT 如何影响 VLM 中的视觉识别能力? Section 5.3: SFT 在 RL 训练中扮演什么角色? Section 5.4: 验证迭代次数如何影响泛化能力? 我们在 GeneralPoints 和 V-IRL 上评估了不同后训练方法的性能, 这两个任务各自包含纯语言 (-L) 和视觉语言 (-VL) 变体, 并且各自涵盖规则变体。对于每个任务, 我们分

别在单一规则上对 RL 和 SFT 的训练计算量进行缩放。我们将训练规则上的结果视为分布内 (ID) 性能, 而在未见规则上的结果则衡量分布外 (OOD) 泛化能力。在 GeneralPoints 中, ID 情况将所有 'J'、'Q'、'K' 视为 10, 而 OOD 情况则将它们解释为 11、12 和 13。至于 V-IRL, ID 情况采用绝对方向坐标系, 而 OOD 情况使用相对方向动作空间。其他细节和额外的实验设置可在 Appendix C 中找到。

RL generalizes, SFT memorizes. 如 Figure 5 所示, RL 在所有任务 (包括单模态 (LLM) 和多模态 (VLM)) 上持续提升 OOD 性能。具体而言, Figure 6 表明 RL 在 GP-L 上实现了 **+3.5%** 的增长 (11.5% $\rightarrow$ 15.0%), 在 V-IRL-L 上实现了 **+11.0%** 的增长 (80.8% $\rightarrow$ 91.8%)。即使在 VLM 中面临视觉识别的额外挑战, RL 仍能保持一致的性能提升, 分别在 GP-VL 上提升 **+3.0%** (11.2% $\rightarrow$ 14.2%),

SFT Memorizes, RL Generalizes

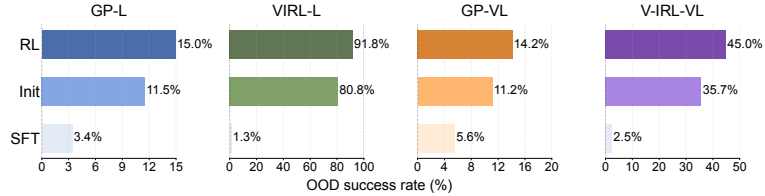


Figure 6: **Comparison of out-of-distribution performance under rule variants.** We report the success rate for GeneralPoints and per-step-accuracy for V-IRL. For each subplot, RL and SFT are trained with equal computation, and their shared initial checkpoint (marked as Init) is set as baseline. Detailed setups are provided in Appendix C.3.

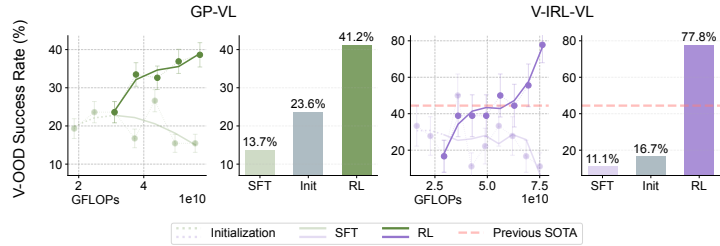


Figure 7: **Comparison of out-of-distribution performance under visual variants.** Similar to Figures 5 and 6, we present both the performance dynamics (shown as lines) and final performance (shown as bars) for visual out-of-distribution evaluations. The previous state-of-the-art on V-IRL VLN mini benchmark (Yang et al., 2024a) is marked in orange. Detailed evaluation setups (and curve smoothing) are provided in Appendix C.3.

consider a natural follow-up question: *How does RL affect VLMs' visual capabilities?* To study this question, we conducted additional ablation studies in the GP-VL environment to investigate the OOD performance of RL and SFT, along with the model's visual recognition accuracy, in terms of recognizing the 4 cards from the input image. In particular, we study how scaling post-training compute via RL/SFT both affects generalization in rule-based OOD (Figure 8 left), and visual recognition accuracy and visual OOD (Figure 8 right).

Scaling RL improves visual recognition accuracy in VLM training. As shown in Figure 8, we observe that the VLMs' visual recognition accuracy largely affects the overall performance, which was similarly observed in Zhong et al. (2024). In addition, scaling up RL compute also improves visual recognition accuracy, as a byproduct of its generalization capability, while scaling SFT deteriorates both visual recognition accuracy and overall performance. Additional experimental results are provided in Figures 16 and 17 of Appendix D.1.

5.4. The Role of SFT for RL Training

Despite the superiority of RL in generalizing the model's reasoning and visual capabilities, as discussed previously, the experimental pipeline still instantiates RL *after* SFT. In this subsection, we focus on another key question: *Is SFT necessary for RL training?* To answer this question, we conduct additional experiments that directly apply end-to-end RL to post-train the base model Llama3.2 using GeneralPoints in the purely language case (Figure 9).

SFT is necessary for RL training when the backbone model does not follow instructions. Figure 9 shows that without SFT, all end-to-end RL runs fail to improve. More specifically, we observe that without SFT, the base model suffers from poor instruction following capability. A detailed failure case is provided in Figure 20 (in Appendix D.3), revealing that the base Llama-3.2-Vision-11B model tends to generate long, tangential, and unstructured responses. This issue makes it impossible to retrieve task-related information and rewards for RL training. Note that due to the difference in backbone model, our results do not contradict with DeepSeekAI et al. (2025), which suggests that SFT is unnecessary for downstream RL training.

SFT Memorizes, RL Generalizes

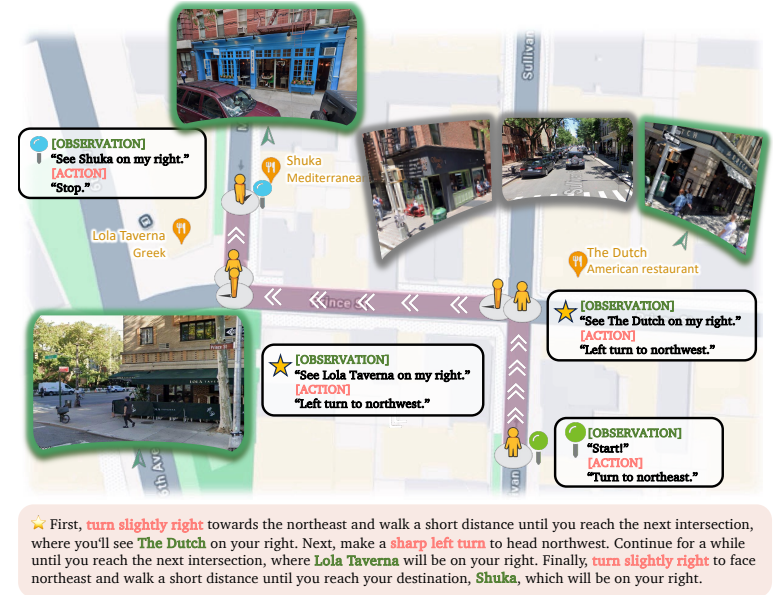


Figure 4: V-IRL 中一项导航任务的演示。智能体在 V-IRL 中遵循给定的语言导航指令在场所间移动。导航过程显示在顶部，导航指令展示在下方。视觉观察相关信息以绿色高亮，动作相关信息以橙色标记。

在 V-IRL-VL 上提升 **+9.3%** (35.7% → 45.0%)。相比之下, SFT 在所有任务的所有 OOD 评估中均表现出性能下降: 在 GP-L 上下降 **-8.1%** (11.5% → 3.4%), 在 V-IRL-L 上下降 **-79.5%** (80.8% → 1.3%), 在 GP-VL 上下降 **-5.6%** (11.2% → 5.6%), 在 V-IRL-VL 上下降 **-33.2%** (35.7% → 2.5%)。

5.1. Generalization in Visual Out-of-Distribution Tasks

第 5 节表明, 强化学习 (RL) 能够在规则变体间实现泛化, 而监督微调 (SFT) 则表现出相反的趋势。由于视觉语言模型 (VLM) 也包含视觉模态, 我们接下来研究视觉变化对分布外 (OOD) 泛化的影响。对于 GeneralPoints, 我们使用黑色花色 (♠, ♣) 训练 VLM, 并在红色花色 (♥, ♦) 上测试其分布外性能。对于 V-IRL, 我们在纽约市收集的路线上

训练模型, 并在包含来自全球不同城市路线的原始 V-IRL VLN 迷你基准 (Yang et al., 2024a) 上对其进行评估 (详见 Appendix B.1)。请注意, 本节实验中的规则保持一致。

RL generalizes in visual OOD tasks. 如 Figure 7 所示, 我们观察到 RL 在视觉 OOD 任务中仍然能够泛化, 而 SFT 则持续表现不佳。具体而言, 在 GP-VL 和 VIRL-VL 中, RL 实现了 **+17.6%** (23.6% → 41.2%) 和 **+61.1%** (16.7% → 77.8%) 的性能提升, 而 SFT 则遭受了 **-9.9%** (23.6% → 13.7%) 和 **-5.6%** (16.7% → 11.1%) 的性能下降。作为这项视觉 OOD 研究的副产品, 我们还表明, 我们的多轮 RL 公式在 V-IRL 迷你基准上将最先进的结果提升了 **+33.8%** (44.0% → 77.8%) (参见 Yang et al. (2024a) 的表 5)。值得注意的是, 与 V-IRL 中报道的先前最先进方法不同 (该方法依赖于两阶段的 VLM-LLM 协作技术以及

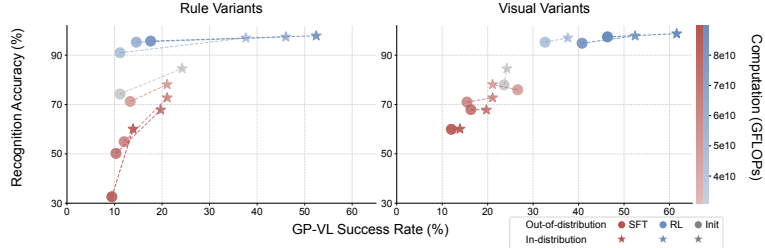


Figure 8: **Recognition vs. success rate for RL and SFT under different variants in GP-VL.** We report both in-distribution (red) and OOD (blue) performance of recognition (y-axis) and episode success rate (x-axis). We denote the training compute of each data point via transparency (color bar) while connected ($\star$ -o) pairs are evaluated using same checkpoints. As scaling up post-training compute, RL improves both recognition and overall accuracy, while SFT shows opposite effect.

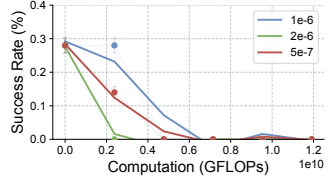


Figure 9: **RL experiments on GP-L without SFT initialization.** All trials fail due to poor instruction following capability of the base model.

5.5. Role of Verification Iterations

Verification serves as another crucial component in our multi-step training and evaluation pipeline (see Figures 2 and 3). To validate its necessity and better understand its effect, we conduct RL experiments with different verification iterations $\{1, 3, 5, 10\}$ using GP-L (Figure 10).

Scaling up verification improves generalization. In Figure 10, we observe that RL generalizes better with more verification steps. More specifically, under the same computational budget across all experiments, we observe improvements of **+2.15%** (3 steps), **+2.99%** (5 steps), **+5.99%** (10 steps). In contrast, in the case with one verification step, we only observe a marginal improvement of **+0.48%** in OOD performance improvement.

6. Conclusion, Discussion, and Limitations

In this paper, we present a comprehensive analysis of the generalization effects of foundation model post-training techniques, specifically RL and SFT. Through extensive

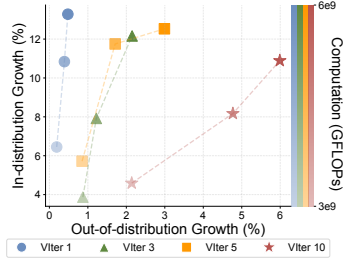


Figure 10: **In-distribution vs. OOD performance growth on GP-L.** We record RL experiments with different number of verification iterations (VIter) as scaling up training compute (color transparency).

experiments on the GeneralPoints and V-IRL tasks, we demonstrated that RL exhibits superior performance in learning generalizable knowledge, while SFT tends to merely memorize the training data, across both the rule and visual variations. This phenomenon consistently occurs across multimodal arithmetic and spatial reasoning capabilities. In addition, we studied the effect of RL on visual recognition, the role of SFT, and the role of verification steps. During our study, two challenges were not resolved.

Failure of SFT on GP-VL. In Figure 5 for GP-VL, we observe that SFT fails to achieve a comparable in-distribution performance with RL. To mitigate the variance introduced by hyperparameter choices, we additionally conduct 10 more experiments with different learning rates and tunable components (Figure 16), none of which exhibits a strong increasing trend like RL (Figure 17).

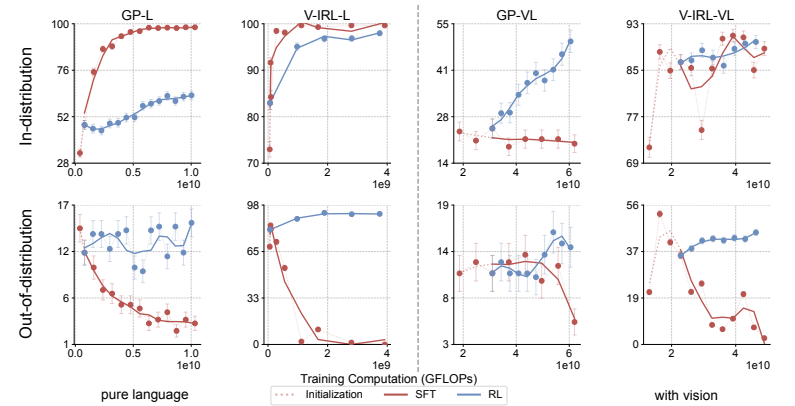


Figure 5: **Success rate (%) - GFLOPs trendlines for RL and SFT on GeneralPoints 与 V-IRL.** 顶行显示分布内性能，底行显示分布外性能。结果针对每个任务的纯语言 (-L) 和视觉语言 (-VL) 变体呈现。对于 GeneralPoints, 我们报告情节成功率, 而对于 V-IRL, 我们报告每步准确率, 其总体成功率见 Figures 1 and 18. 详细评估设置 (及曲线平滑) 见 Appendix C.3.

对闭源模型 (OpenAI, 2023a) 的定制提示工程), 我们的端到端 RL 方法使开源模型 (Dubey et al., 2024) 达到了更优的性能。

5.2. RL Improves Visual Capabilities

基于上述观察, 即通过 RL 训练的 VLM 能够泛化到视觉 OOD 任务 (第 5.1 节), 我们考虑一个自然的后续问题: RL 如何影响 VLM 的视觉能力? 为了研究这个问题, 我们在 GP-VL 环境中进行了额外的消融研究, 以调查 RL 和 SFT 的 OOD 性能, 以及模型在从输入图像中识别 4 张牌方面的视觉识别准确率。具体而言, 我们研究了通过 RL/SFT 扩展后训练计算量如何影响基于规则的 OOD 泛化 (Figure 8 左图), 以及视觉识别准确率和视觉 OOD (Figure 8 右图)。

Scaling RL improves visual recognition accuracy in VLM training. 如 Figure 8 所示, 我们观察到 VLM 的视觉识别准确率在很大程度上影响整体性能, 这与 Zhong et al. (2024) 中的观察类似。此外, 增加 RL 计算量也提高了视觉识别准确率, 这是其泛化能力

的副产品, 而增加 SFT 计算量则会降低视觉识别准确率和整体性能。更多实验结果见附录 D.1 的 Figures 16 and 17。

5.3. The Role of SFT for RL Training

尽管如前所述, 强化学习在泛化模型的推理和视觉能力方面具有优越性, 但实验流程仍然是在监督微调之后实例化强化学习。在本小节中, 我们关注另一个关键问题: 监督微调对于强化学习训练是否必要? 为了回答这个问题, 我们进行了额外的实验, 在纯语言场景下 (Figure 9), 直接应用端到端强化学习, 使用 GeneralPoints 对基础模型 Llama3.2 进行后训练。

SFT is necessary for RL training when the backbone model does not follow instructions. Figure 9 显示, 在没有监督微调的情况下, 所有端到端强化学习运行均未能取得改进。更具体地说, 我们观察到, 在没有监督微调的情况下, 基础模型的指令遵循能力很差。Figure 20 (位于 Appendix D.2 中) 提供了一个详细的失败案例, 揭示了基础 Llama3.2-Vision-11B

Given our observation that scaling up SFT degrades visual recognition capabilities (Figure 8), we hypothesize that SFT locally overfits to reasoning tokens while neglecting recognition tokens, possibly due to the higher frequency of reasoning tokens (see Figure 11 as example). We leave further investigation to future work.

Limits of RL in corner cases. As discussed in Section 5.4, SFT is necessary for effective RL training on Llama-3.2. We investigate applying RL to an overly-tuned SFT checkpoint. As demonstrated in Figure 19, RL is unable to recover out-of-distribution performance when starting from such a checkpoint. Example failure cases are illustrated in Figure 21, where the model collapses to the training rule. These results, together with findings in Section 5.4, indicate that RL has limited effectiveness when applied to extremely underfit or overfit initial checkpoints. Further research is needed to delineate the conditions under which SFT facilitates effective RL.

Impact Statement

This paper presents work aimed at advancing the field of Machine Learning. While the study includes tasks such as GeneralPoints, which is a synthetic environment, and V-IRL, a real-world map simulator, our work is confined to controlled research settings. The V-IRL environment is designed as a simulated proxy for real-world tasks, but no deployment or interaction with actual real-world systems or data was involved. The methods, environments, and tasks investigated in this study were constructed to advance our understanding of model generalization without introducing any foreseeable societal or ethical implications.

Acknowledgements

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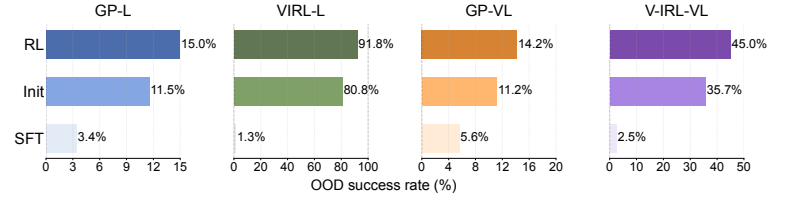


Figure 6: **Comparison of out-of-distribution performance under rule variants.** 我们报告 GeneralPoints 的成功率以及 V-IRL 的 per-step-accuracy。对于每个子图，RL 和 SFT 使用等量计算进行训练，其共享的初始检查点（标记为 Init）设为基线。详细设置见 Appendix C.3。

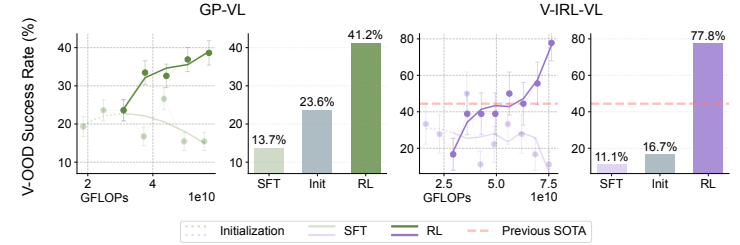


Figure 7: **Comparison of out-of-distribution performance under visual variants.** 与 Figures 5 和 6 类似，我们展示了视觉分布外评估的性能动态（以线条表示）和最终性能（以条形图表示）。V-IRL VLN 小型基准 (Yang et al., 2024a) 上先前的最先进水平以 橙色 标出。详细的评估设置（及曲线平滑处理）在 Appendix C.3 中提供。

模型倾向于生成长篇、离题且无结构的回答。这个问题使得无法为强化学习训练检索任务相关的信息和奖励。请注意，由于骨干模型的差异，我们的结果与 DeepSeekAI et al. (2025) 并不矛盾，后者认为监督微调对于下游强化学习训练是不必要的。

5.4. Role of Verification Iterations

验证是我们多步骤训练和评估流程中的另一个关键组成部分（参见 Figures 2 和 3）。为了验证其必要性并更好地理解其效果，我们使用 GP-L (Figure 10) 进行了不同验证迭代次数 {1, 3, 5, 10} 的强化学习实验。

Scaling up verification improves generalization. 在 Figure 10 中，我们观察到随着验证步骤的增加，强化学习的泛化能力更好。更具体地说，在所有实验保

持相同计算预算的情况下，我们观察到性能分别提升了 +2.15%（3 步）、+2.99%（5 步）、+5.99%（10 步）。相比之下，在只有一个验证步骤的情况下，我们仅观察到分布外性能有 +0.48% 的边际提升。

6. Conclusion, Discussion, and Limitations

本文中，我们对基础模型后训练技术（特别是 RL 和 SFT）的泛化效应进行了全面分析。通过在 GeneralPoints 和 V-IRL 任务上的大量实验，我们证明 RL 在学习可泛化知识方面表现出更优的性能，而 SFT 倾向于仅仅记忆训练数据，这一现象在规则和视觉变体上均成立。该规律在多模态算术与空间推理能力中持续出现。此外，我们研究了 RL 对视觉识别的影响、SFT 的作用以及验证步骤的角色。在研究过程中，有两个挑战尚未解决。

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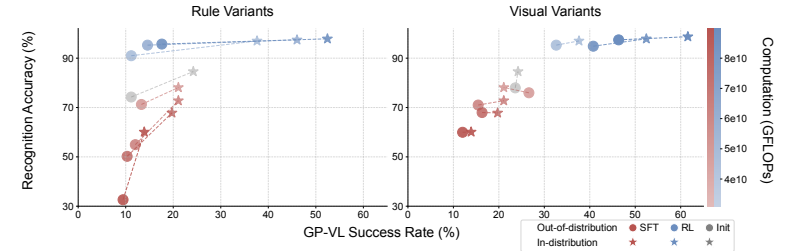


Figure 8: **Recognition vs. success rate for RL and SFT under different variants in GP-VL.** 我们报告了识别 (y 轴) 和回合成功率 (x 轴) 的分布内 (红色) 和 OOD (蓝色) 性能。我们通过透明度 (颜色条) 表示每个数据点的训练计算量, 而相连的 (x-o) 数据点对使用相同的检查点进行评估。随着后训练计算量的增加, RL 同时提升了识别能力和整体准确率, 而 SFT 则显示出相反的效果。

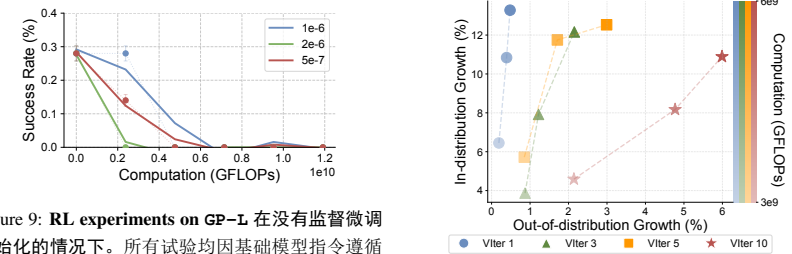


Figure 9: **RL experiments on GP-L 在 没有监督微调初始化的情况下。** 所有试验均因基础模型指令遵循能力差而失败。

Failure of SFT on GP-VL. 在 Figure 5 中针对 GP-VL 的实验中, 我们观察到 SFT 未能达到与 RL 相当的分布内性能。为减少超参数选择引入的方差, 我们额外进行了 10 组不同学习率和可调组件 (Figure 16) 的实验, 其中没有任何一组像 RL 那样呈现明显的上升趋势 (Figure 17)。鉴于我们观察到扩大 SFT 规模会损害视觉识别能力 (Figure 8), 我们假设 SFT 在局部过度拟合推理标记而忽略了识别标记, 这可能是由于推理标记出现频率更高所致 (示例参见 Figure 11)。我们将进一步研究留待未来工作。

Limits of RL in corner cases. 如 Section 5.3 所述, SFT 对于 Llama-3.2 的有效 RL 训练是必要的。我们研究了将 RL 应用于过度调优的 SFT 检查点的情况。

Figure 10: **In-distribution vs. OOD performance growth on GP-L.** 我们记录了具有不同验证迭代次数 (Viter) 的强化学习实验, 随着训练计算量的增加 (颜色透明度)。

如 Figure 19 所示, 从这类检查点开始训练时, RL 无法恢复分布外性能。Figure 21 展示了示例失败案例, 其中模型坍塌至训练规则。这些结果与 Section 5.3 中的发现共同表明, 当应用于极端欠拟合或过拟合的初始检查点时, RL 的效果有限。需要进一步研究以明确 SFT 促进有效 RL 训练的条件范围。

Impact Statement

本文旨在推动机器学习领域的发展。虽然研究包含了诸如 GeneralPoints (一种合成环境) 和

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. [3](#), [18](#)

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V-IRL (一种真实世界地图模拟器) 等任务, 但我们的工作仅限于受控的研究环境。V-IRL 环境被设计为真实世界任务的模拟代理, 但并未涉及任何实际真实世界系统或数据的部署或交互。本研究探讨的方法、环境和任务旨在增进我们对模型泛化能力的理解, 而未引入任何可预见的社会或伦理影响。

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A. Details on the General Points Environment

In this section, we demonstrate the design details for `GeneralPoints` mentioned in Section 4.1. We first present the data used for this environment (Appendix A.1). Then, we show examples of the environment’s transition dynamics (Appendix A.2), followed by a description of key arguments and reward design specification (Appendix A.3).

A.1. Data

`GeneralPoints` card quadruples are sampled from a deck of 52 standard poker cards. Each sampled quadruple is guaranteed to have at least one solution equals the target point, i.e. 24. We ensure this by using an expert solver during the sampling process.

A.2. Detailed Examples on the Transition Dynamics

As shown in Figure 11 and Figure 12, we treat the system prompt as v_0^{in} and then subsequently appending the future outputs $v_{1:t}^{\text{out}}$ and verifier info $v_{1:t}^{\text{ver}}$ into the prompt for getting the $t + 1$ output. Figure 11 provides an example with the visual inputs, while Figure 12 shows the language only case.

A.3. Additional Eetails on the Environmental Design

Arguments. The `GeneralPoints` environment supports the following configurable arguments:

- Target point: Any positive integer
- Face cards rule: Two options
 - ‘J’, ‘Q’, and ‘K’ all count as ‘10’
 - ‘J’, ‘Q’, and ‘K’ count as ‘11’, ‘12’, and ‘13’ respectively
- Card sampling: Two options
 - Sample 4 cards without replacement from a deck of 52 poker cards
 - Sample at least one card from ‘J’, ‘Q’, and ‘K’
- Card color: Three options
 - Black suits only: ♠, ♣.
 - Red suits only: ♥, ♦.
 - All suits: ♠, ♥, ♣, ♦.

For all experiments, we fix the target point at 24. In Figure 5, training and in-domain evaluation use the rule where face cards count as ‘10’. For out-of-domain evaluation, we use the alternative face cards rule and require at least one face card, forcing calculations with numbers above 10 that are not encountered during training. For visual distribution

shift experiments (Section 5.2), we train the model on black suits ♠, ♣ and evaluate out-of-domain performance on red suits ♥, ♦.

Reward design. An episode terminates when either a correct equation is generated or the maximum verification step of 5 is reached. The reward function is as follows:

- $r = 5$: For generating a legal equation that equals the target point
- $r = -1$: For legal equations using each card once but not equaling the target point
- $r = -1$: For exceeding maximum verification step
- $r = -2$: For legal equations containing numbers not among the given choices
- $r = -3$: For all other illegal equations

In the vision-language variant (`GeneralPoints-VL`), an additional penalty of $r = -1.5$ is applied when the agent fails to correctly recognize the given cards.

B. Details on the V-IRL Environment

Similar to Appendix A, we present the design details for V-IRL discussed in Section 4.2. First, we introduce the database used for this environment (Appendix B.1) and demonstrate transition examples (Appendix B.2). We then describe the environment by explaining its fundamental component—*route*. Finally, we outline our modifications and reward design choices made to adapt the original V-IRL for reinforcement learning training (Appendix B.3).

B.1. Data

Leveraging the data collection pipeline of Yang et al. (2024a), we construct a training database with 1000 unique routes from New York City. We evaluate all rule-variant experiments and visual in-distribution experiments using randomly sampled routes from this database. For visual out-of-distribution experiments, we directly adopt the VLN mini benchmark from Yang et al. (2024a). This benchmark consists of 18 distinct routes across nine cities: Milan, New Delhi, Buenos Aires, London, Hong Kong, New York,⁴ Melbourne, Lagos, and San Francisco, with two routes per city.

⁴These NYC routes in the VLN mini benchmark do not overlap with our training data.

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System Prompt (v_0^{in})**[Task Description]**

You are an expert 24 points card game player. You are observing these four cards in the image. Note that 'J', 'Q', and 'K' count as '10', and each card must be used once. Your goal is to output a formula that evaluates to 24 using numbers from the cards and operators such as '+', '-', '*', '/', '(', ')', and '='.

[Output]

Your response should be a valid json file in the following format:

```
{
  "cards": [x, y, z, w], where 'J', 'Q', and 'K' count as '10',
  "number": [a, b, c, d], where a, b, c, and d are the numbers on the cards,
  "formula": "an equation that equals 24",
}
```

**Appending model and verifier outputs to obtain v_t^{in}**

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$
Model output (v_t^{out})

```
{
  "cards": ['A', '3', 'K', '6'],
  "number": [1, 3, 13, 6],
  "formula": "(1+6)*3+13=24",
}
```

Verifier Output (v_t^{ver})

You failed this trial because your formula is incorrect.

$$\triangleright v_{t+1}^{\text{in}} = \text{concat}(v_t^{\text{in}}, v_t^{\text{out}}, v_t^{\text{ver}})$$

Figure 11: **An example of our prompt update** for constructing v_{t+1}^{in} using v_t^{in} , v_t^{out} and v_t^{ver} . This example provides an optional vision input for VLMs, adding a visual recognition challenge. The brown parts marks the task and related information, and the purple parts denote the state (s_t) specific info. The blue and red describe the output from the model and verifier, respectively.

B.2. Detailed Examples on the Transition Dynamics

We provide detailed transition examples of the V-IRL environment in Figure 13 (vision and language) and Figure 14 (pure language).

B.3. Additional Details on the Environmental Design

Concept of route. The route serves as the fundamental navigation object in the V-IRL environment. As illustrated in Figure 4, each route corresponds to a real-world path with associated language instructions and visual signals. Using Figure 4 as an example, a route comprises:

- Destination: Shuka
- Starting point: Start
- Turning points: The Dutch, Lola Taverna
- Straight road: Roads connecting turning points, starting point, and destination
- Street views: 360-degree panoramic views at each movable point

- Oracle information: Expert observation data for each movable point
- Expert trajectory
- Instruction

Although the instructions in Figures 4, 13 and 14 are presented in different formats, they convey equivalent information, with Figure 4 using natural language.

Simplification and arguments. We simplify the original V-IRL design from Yang et al. (2024a) to better accommodate RL training. The modifications include eliminating the 2-stage navigation pipeline that required a separate visual detector for street view processing, and removing online queries to reduce training time and cost. Our V-IRL environment contains 2 additional configuration arguments compared with the original design:

- Action space: two options
 - Absolute direction: "turn_direction(x)" where $x \in \{\text{'north'}, \text{'northeast'}, \text{'south'}, \text{'southeast'}, \text{'west'}, \text{'northwest'}, \text{'southwest'}, \text{'south-southwest'}, \text{'south-southeast'}, \text{'north-northwest'}, \text{'north-northeast'}\}$

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System Prompt (v_0^{in})

[Task Description]
You are an expert 24 points card game player. You are observing these four cards in the image. Note that 'J', 'Q', and 'K' count as '11', '12', and '13' respectively, and each card must be used once. Your goal is to output a formula that evaluates to 24 using numbers from the cards and operators such as '+', '-', '*', '/', '(', ')', and '='.

[Input]
Cards: ['A', '3', 'K', '6']

[Output]
Your response should be a valid json file in the following format:
{
 "cards": [x, y, z, w], where 'J', 'Q', and 'K' count as '10',
 "number": [a, b, c, d], where a, b, c, and d are the numbers on the cards,
 "formula": "an equation that equals 24",
}

Appending model and verifier outputs to obtain v_t^{in}

$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{cr}}, v_1^{\text{out}}, v_1^{\text{cr}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{cr}}]$ $\triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{cr}}]_{k=0}^{t-1})$

Model output (v_t^{out})

{
 "cards": ['A', '3', 'K', '6'],
 "number": [1, 3, 13, 6],
 "formula": "(1+6)*3+13=24",
}

Verifier Output (v_t^{cr})

You failed this trial because your formula is incorrect. $\triangleright v_{t+1}^{\text{in}} = \text{concat}(v_t^{\text{in}}, v_t^{\text{out}}, v_t^{\text{cr}})$

Figure 12: An example of our prompt update for constructing v_{t+1}^{in} using v_t^{in} , v_t^{out} and v_t^{cr} . This example provides an optional vision input for VLMs, adding a visual recognition challenge. The brown parts marks the task and related information, and the purple parts denote the state (s_t) specific info. The blue and red describe the output from the model and verifier, respectively.

'east', 'southeast', 'south', 'southwest', 'west', 'northwest', 'forward()', "stop()"

- Relative direction:
"turn_direction(x)" where $x \in \{\text{'left', 'right', 'slightly left', 'slightly right', 'forward()', 'stop()'}\}$
- Maximum straight road length: any positive integer

The action space argument accommodates the rule variants described in Section 4. For experiments shown in Figure 5, we use absolute direction action space during training and in-domain evaluation, while using the alternative rule for out-of-domain evaluation. We implement a maximum straight road length to limit the number of movable coordinates between turning points, preventing sequences of repetitive "forward()" actions. We conduct visual distribution shift experiments (Section 5.2) via training the model on New York City regions and evaluating the out-of-domain performance on the worldwide navigation routes from the benchmark released by Yang et al. (2024a).

Reward design. An episode terminates when either the navigation agent stops at the destination or the maximum verification step of 2 is reached. The reward function is as follows:

- $r = 1$: For generating a correct action at the current coordinate
- $r = -1$: For generating wrong action at the current coordinate
- $r = -1$: For exceeding maximum verification step
- $r = -1.5$: For failed detection of landmarks

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System Prompt (v_0^{in})**[Task Description]**

You are an expert in navigation. You will receive a sequence of instructions to follow while observing your surrounding street views. You are also provided with your observation and action history in text. your goal is to take the action based on the current observation and instruction.

[Instruction]

1. First, turn left to face east.
2. Move forward until you reach the next intersection where Hotel 32One is on your right behind.
3. Turn right to face north.
4. Move forward until you reach the next intersection where Dragon Gate Chinatown SF is on your right front.
5. Turn left to face east.
6. Move forward until the destination Café de la Presse is on your right.

[Current observation]

You observe a 2x2 grid of street view images with the following headings:

[front, right
back, left]

You need to identify if any of the landmarks in the instruction are visible in the street view grid.

[Action space]

- "forward()": indicates moving forward for 1 step;
- "turn_direction(x)": indicates turn direction to the target heading, where $x \in \text{'north', 'northeast', 'east', 'southeast', 'south', 'southwest', 'west', 'northwest'}$;
- "stop()": indicates the navigation is finished;

[Observations and actions sequence]

O_1: No landmarks nearby;

A_1: turn_direction(east)

O_2: No landmarks nearby;

A_2: forward()

O_3: No landmarks nearby;

A_3: forward()

O_4: You observe an image of 4 views; You observe an intersection

A_4:

[Output]

Your response should be a valid json file in the following format:

```
{
  "current observation": latest observation from the street view grid,
  "current instruction": analyze the full instruction and identify the sentence to be executed,
  "action": the action to be taken chosen from the action space,
}
```

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

Model output (v_t^{out})

```
{
  "current observation": "Hotel 32One is on my right behind; I observe an intersection",
  "current instruction": "Turn right to face north.",
  "action": "turn_direction(north)",
}
```

Verifier Output (v_t^{ver})

Correct solution.

$\triangleright$ Move to the next scene upon correct action



Figure 13: An example of our prompt update for constructing v_{t+1}^{in} using v_t^{in} , v_t^{out} and v_t^{ver} . This example provides an optional vision input for VLMs, adding a visual recognition challenge. The brown parts marks the task and related information, and the purple parts denote the state (s_t) specific info. The blue and red describe the output from the model and verifier, respectively.

A. Details on the General Points Environment

在本节中，我们将详细阐述 Section 4.1中提及的 GeneralPoints 环境设计细节。首先介绍用于该环境的数据 (Appendix A.1)。接着，展示环境状态转移动态的示例 (Appendix A.2)，随后说明关键参数与奖励函数的设计规范 (Appendix A.3)。

A.1. Data

GeneralPoints 的四张牌组合是从一副 52 张的标准扑克牌中抽取的。每次抽取的四张牌组合都保证至少存在一种解法，其计算结果等于目标点数，即 24。我们通过抽样过程中使用专家求解器来确保这一点。

A.2. Detailed Examples on the Transition Dynamics

如 Figure 11 和 Figure 12 所示，我们将系统提示视为 v_0^{in} ，随后将未来的输出 $v_{1:t}^{\text{out}}$ 和验证器信息 $v_{1:t}^{\text{ver}}$ 附加到提示中，以获取 $t+1$ 输出。Figure 11 提供了一个包含视觉输入的示例，而 Figure 12 则展示了纯语言的情况。

A.3. Additional Eetails on the Environmental Design

Arguments. GeneralPoints 环境支持以下可配置参数：

- 目标点数：任意正整数
- 人头牌规则：两个选项
 - 'J'、'Q' 和 'K' 均计为 '10'
 - 'J'、'Q' 和 'K' 分别计为 '11'、'12' 和 '13'
- 卡牌抽样：两个选项
 - 从一副 52 张的扑克牌中无放回地抽取 4 张牌
 - 从 'J'、'Q' 和 'K' 中至少抽取一张牌
- 卡牌花色：三个选项
 - 仅黑桃花色：♠, ♠, ♠.
 - 仅红桃花色：♥, ♥, ♥.

- 所有花色：♠, ♥, ♣, ♦.

在所有实验中，我们将目标点数固定为 24。在 Figure 5 中，训练和域内评估使用人头牌计为 '10' 的规则。对于域外评估，我们使用替代的人头牌规则并要求至少包含一张人头牌，这迫使模型进行训练中未遇到的、涉及大于 10 的数字的计算。对于视觉分布偏移实验 (Section 5.1)，我们在黑色花色 ♠, ♣ 上训练模型，并在红色花色 ♥, ♦ 上评估其域外性能。

Reward design. 当生成正确的等式或达到最大验证步数 5 时，一个回合终止。奖励函数如下：

- $r = 5$ ：用于生成等于目标点数的合法等式
- $r = -1$ ：用于合法使用每张牌一次但结果不等于目标点数的等式
- $r = -1$ ：用于超过最大验证步数
- $r = -2$ ：用于包含非给定数字的合法等式
- $r = -3$ ：用于所有其他非法等式

在视觉-语言变体 (GeneralPoints-VL) 中，当智能体未能正确识别给定的卡牌时，会额外施加 $r = -1.5$ 的惩罚。

B. Details on the V-IRL Environment

类似于 Appendix A，我们介绍在 Section 4.2 中讨论的 V-IRL 的设计细节。首先，我们介绍用于此环境的数据库 (Appendix B.1) 并展示状态转移示例 (Appendix B.2)。接着，我们通过解释其基本组件——route 来描述该环境。最后，我们概述了为将原始 V-IRL 适配于强化学习训练 (Appendix B.3) 而进行的修改和奖励设计选择。

B.1. Data

利用 Yang et al. (2024a) 的数据收集流程，我们构建了一个包含 1000 条纽约市独特路线的训练数据库。我们使用从该数据库中随机采样的路线来评估所有规则变体实验和视觉域内实验。对于视觉域外实验，

System Prompt (v_0^{in})**[Task Description]**

You are an expert in navigation. You will receive a sequence of instructions to follow. You are also provided with your observation and action history in text. Your goal is to first analyze the instruction and identify the next sentence to be executed. Then, you need to provide the action to be taken based on the current observation and instruction.

[Instruction]

1. First, turn left to face east.
2. Move forward until you reach the next intersection where Hotel 32One is on your right behind.
3. Turn right to face north.
4. Move forward until you reach the next intersection where Dragon Gate Chinatown SF is on your right front.
5. Turn left to face east.
6. Move forward until the destination Café de la Presse is on your right.

[Action space]

- "forward()": indicates moving forward for 1 step;
- "turn_direction(x)": indicates turn direction to the target heading, where $x \in \{\text{'north', 'northeast', 'east', 'southeast', 'south', 'southwest', 'west', 'northwest'}\}$;
- "stop()": indicates the navigation is finished;

[Observations and actions sequence]

O_1: No landmarks nearby;
A_1: turn_direction(east)
O_2: No landmarks nearby;
A_2: forward()
O_3: No landmarks nearby;
A_3: forward()
O_4: Hotel 32One is on your right behind; You observe an intersection
A_4:

[Output]

Your response should be a valid json file in the following format:

```
{
  "current observation": latest observation from the street view grid,
  "current instruction": analyze the full instruction and identify the sentence to be executed,
  "action": the action to be taken chosen from the action space,
}
```

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \quad \triangleright \quad v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

Model output (v_t^{out})

```
{
  "current observation": "Hotel 32One is on my right behind; I observe an intersection",
  "current instruction": "Turn right to face north.",
  "action": "turn_direction(north)",
}
```

Verifier Output (v_t^{ver})

Correct solution. $\triangleright$ Move to the next scene upon correct action

Figure 14: An example of our prompt update for constructing v_{t+1}^{in} using v_t^{in} , v_t^{out} and v_t^{ver} . The brown parts marks the task and related information, and the purple parts denote the state (s_t) specific info. The brown parts marks the task and related information, and the purple parts denote the state (s_t) specific info. The blue and red describe the output from the model and verifier, respectively.

System Prompt (v_0^{in})**[Task Description]**

You are an expert 24 points card game player. You are observing these four cards in the image. Note that 'J', 'Q', and 'K' count as '10', and each card must be used once. Your goal is to output a formula that evaluates to 24 using numbers from the cards and operators such as '+', '-', '*', '/', '(', ')', and '='.

**[Output]**

Your response should be a valid json file in the following format:

```
{
  "cards": [x, y, z, w], where 'J', 'Q', and 'K' count as '10',
  "number": [a, b, c, d], where a, b, c, and d are the numbers on the cards,
  "formula": "an equation that equals 24",
}
```

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \quad \triangleright \quad v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

Model output (v_t^{out})

```
{
  "cards": ['A', '3', 'K', '6'],
  "number": [1, 3, 13, 6],
  "formula": "(1+6)*3+13=24",
}
```

Verifier Output (v_t^{ver})

You failed this trial because your formula is incorrect.

$$\triangleright v_{t+1}^{\text{in}} = \text{concat}(v_t^{\text{in}}, v_t^{\text{out}}, v_t^{\text{ver}})$$

Figure 11: 我们提示更新的一个示例，用于使用 v_{t+1}^{in} 和 v_t^{in} , v_t^{out} 构建 v_t^{ver} 。此示例为视觉语言模型提供了可选的视觉输入，增加了视觉识别挑战。其中棕色部分标记任务及相关信息，紫色部分表示状态 (s_t) 特定信息。蓝色和红色分别描述来自模型和验证器的输出。

我们直接采用来自 Yang et al. (2024a) 的 VLN mini 基准测试。该基准测试包含跨越九个城市的 18 条不同路线：米兰、新德里、布宜诺斯艾利斯、伦敦、香港、纽约、⁴ 墨尔本、拉各斯和旧金山，每个城市两条路线。

B.2. Detailed Examples on the Transition Dynamics

我们提供了 V-IRL 环境在 Figure 13（视觉与语言）和 Figure 14（纯语言）方面的详细转换示例。

⁴VLN mini 基准测试中的这些纽约市路线与我们的训练数据没有重叠。

B.3. Additional Details on the Environmental Design

Concept of route. 路线是 V-IRL 环境中的基本导航对象。如 Figure 4 所示，每条路线都对应一条现实世界中的路径，并伴有相应的语言指令和视觉信号。以 Figure 4 为例，一条路线包含：

- 目的地：Shuka
- 起点：Start
- 转向点：The Dutch, Lola Taverna
- 直行路段：连接转向点、起点和目的地的道路

C. Experimental Setup

This section details the experimental setup used in Section 5. We first describe our data collection setup for supervised fine-tuning (Appendix C.1). Then, we present the training pipeline (Appendix C.2). Finally, we describe our evaluation metrics and the statistical tools used for generating plots (Appendix C.3).

C.1. Data

SFT data collection. As illustrated in Figures 11 to 14, GeneralPoints and V-IRL environments naturally align with prompt-response dialogue structures. We create training samples by pairing each system prompt with its corresponding expert response. All SFT experiments in the main body use optimal single-turn prompt-response pairs, without any verification or revision steps.

SFT on sub-optimal trajectories To examine how more diverse SFT data affects the out-of-distribution performance of SFT, we conduct an ablation study on GP-L using sub-optimal trajectories as training data. Unlike expert prompt-response pairs, these sub-optimal trajectories include errors and verification messages in their prompts. This format aligns with evaluation scenarios where multiple verification iterations are allowed, similar to the data being used for the downstream RL training. In Figure 15, we observe that SFT still merely memorizes the training data with degraded out-of-distribution performance. This evidence suggests that memorization occurs due to the fundamental nature of SFT training rather than the SFT data.

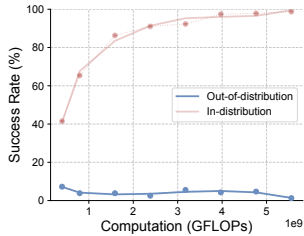


Figure 15: **SFT experiments on GP-L with suboptimal trajectories.** Similar to results in Figure 5, SFT overfits the training data even we increase the trajectory diversity.

C.2. Training Pipeline

As illustrated in Section 5, we follow the training pipeline by RL4VLM (Zhai et al., 2024a), where we first initial-

ize the model with SFT, then separately scale up the compute for SFT and RL (Schulman et al., 2017), starting from this initialized model. For all experiments of SFT and RL in the main body, we tune all components using a shared learning rate per experiment. All training experiments are conducted on an 8 H800 machine (80GB).

C.3. Evaluation Metric

Per-step accuracy. We report the per-step accuracy for V-IRL-VL task in Figures 5 and 6. An individual step is considered correct when the model’s chosen action matches the expert trajectory at that position. Note that intermediate verification steps are counted as independent samples here.

Success rate. We report the success rate (%) of GP-L, GP-VL, V-IRL-L and V-IRL-VL in Figures 5 and 6. In the GeneralPoints task, success is defined as succeeding at least once during the inference time verification. In the V-IRL task, a sample is recorded as success when the model takes correct action at each movable point on the route.

Computation estimation. We estimate the FLOPs for training X following the similar manner of (Snell et al., 2024; Hoffmann et al., 2023), where $X_{train} = 6ND_{train}$ and $X_{inference} = 2ND_{inference}$. Here, N represents the model parameters and D_{train} represents the number of tokens during training. Suppose our SFT and RL experiments start from a checkpoint trained on D_{init} tokens, we can estimate the training computation of SFT and RL via the following equations:

$$\begin{aligned} X_{SFT} &= 6N(D_{init} + D_{SFT}) \\ X_{RL} &= 6N(D_{init} + D_{RL}) + 2ND_{buffer} \end{aligned}$$

Note that the used on-policy RL algorithm PPO (Schulman et al., 2017) contains iterative stages of replay buffer collection and optimization, hence requiring additional inference computation. For simplicity, we approximate the term via:

$$\begin{aligned} D_{buffer} &\approx \frac{E\bar{d}_i\bar{d}_o}{D_{RL}} \cdot D_{RL} \\ &= \lambda D_{RL} \end{aligned}$$

where $E \in \mathbb{N}$ denotes the number of auto-regressive generation processes, $\bar{d}_i, \bar{d}_o$ denote average input tokens and output tokens. We estimate the λ for GeneralPoints and V-IRL as 6 and 5.1 respectively after calculation.

Line smoothing and error bar. All line plots in our paper adopt Savitzky-Golay filter with polynomial order 3 as smoothing function. We assume each evaluated data point

System Prompt (v_0^{in})

[任务描述] 你是一位精通24点纸牌游戏的专家。你正在观察图像中的这四张牌。请注意，'J'、'Q' 和 'K' 分别计为'11'、'12' 和 '13'，且每张牌必须使用一次。你的目标是输出一个使用牌面数字以及运算符如'+', '-', '*', '/', '(', 'y' 和 '='，计算结果为24的公式。

[输入]

牌面: ['A', '3', 'K', '6']

[输出]

你的响应应是一个符合以下格式的有效 json 文件:

```
{
  "cards": [x, y, z, w], 其中'J'、'Q' 和 'K' 计为'10',
  "number": [a, b, c, d], 其中 a, b, c, d 是牌面上的数字,
  "formula": " 一个等于24的等式",
}
```

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \quad \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

Model output (v_t^{out})

```
{
  "cards": ['A', '3', 'K', '6'],
  "number": [1, 3, 13, 6],
  "formula": "(1+6)*3+13=24",
}
```

Verifier Output (v_t^{ver})

你本次尝试失败，因为你的公式不正确。

$$\triangleright v_{t+1}^{\text{in}} = \text{concat}(v_t^{\text{in}}, v_t^{\text{out}}, v_t^{\text{ver}})$$

Figure 12: 我们提示更新的一个示例，用于使用 v_{t+1}^{in} 和 $v_t^{\text{in}}, v_t^{\text{out}}$ 构建 v_t^{ver} 。此示例为视觉语言模型提供了可选的视觉输入，增加了视觉识别挑战。其中棕色部分标记任务及相关信息，紫色部分表示状态 (s_t) 特定信息。蓝色和红色分别描述来自模型和验证器的输出。

- 街景：每个可移动点处的 360 度全景视图
- 先知信息：每个可移动点处的专家观察数据
- 专家轨迹
- 指令

尽管 Figures 4, 13 and 14 中的指令以不同格式呈现，但它们传递了等价的信息，其中 Figure 4 使用了自然语言。

Simplification and arguments. 我们对 Yang et al. (2024a) 中的原始 V-IRL 设计进行了简化，以更好

地适应强化学习训练。修改包括：取消了需要独立视觉检测器处理街景的两阶段导航流程，并移除了在线查询以减少训练时间和成本。与原始设计相比，我们的 V-IRL 环境包含了 2 个额外的配置参数：

- 动作空间：两个选项
 - 绝对方向：

"turn_direction(x)", 其中 $x \in \{\text{'north', 'northeast', 'east', 'southeast', 'south', 'southwest', 'west', 'northwest'}, \text{'forward()'}, \text{'stop()'}\}$
 - 相对方向：

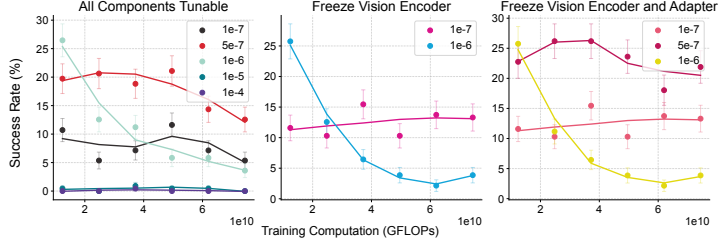


Figure 16: **Ablation studies on GeneralPoints-VL SFT.** We ablate the learning rate and report the in-distribution episode success rate (%) of all experiments. None of the experiments shows an increasing trend beyond 30% success rate.

follows a binomial distribution and approximate the standard error using $\sqrt{\frac{P(1-P)}{N}}$, where P is the demical success rate and N is the number of samples.

D. Additional Experimental Results

In this section, we provide additional experimental results that are not covered in the main body.

D.1. Ablation Studies on GP-VL

As mentioned in Section 6, we observe an abnormal phenomenon that SFT fails to achieve comparable in-distribution performance with RL (see Figure 5 subplot row 1 column 3). To further explore this, we conduct ablation studies over different hyperparameter choices.

SFT. We ablate the hyperparameter choices under the same task setting of GP-VL in Section 5.1. For experiments fine-tuning all parameters, we search learning rates from $\{1 \times 10^{-4}, 1 \times 10^{-4}, 1 \times 10^{-5}, 1 \times 10^{-6}, 5 \times 10^{-7}, 1 \times 10^{-7}\}$. Freezing the vision encoder, we search learning rates $\{1 \times 10^{-6}, 1 \times 10^{-7}\}$. Freezing vision encoder and adapter, we search learning rates $\{1 \times 10^{-6}, 5 \times 10^{-7}, 1 \times 10^{-7}\}$. We provide the in-distribution success rate curve in Figure 16.

RL. Finding suitable hyperparameters for RL experiments requires minimal effort. We conduct a search over learning rates $2 \times 10^{-6}, 1 \times 10^{-6}$, with the in-distribution success rate curves shown in Figure 17. All parameters are tunable in our RL experiments.

D.2. More results on V-IRL-VL

Echoing per-step accuracy results in Figure 5, we report the overall success rate of V-IRL-VL in Figure 18. Due to the task's complexity, both training methods achieve overall success rates no higher than 1%. For V-IRL, the overall

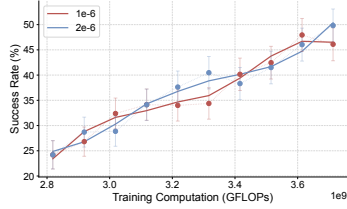


Figure 17: **Ablation studies on GeneralPoints-VL RL.** Echoing Figure 16, we ablate the learning rate and report the in-distribution episode success rate (%) of the two experiments. All components are tunable here.

success rate is a significantly more demanding metric since it aggregates per-step errors. For example, a random policy achieving 10% per-step accuracy would achieve only approximately $10^{-8}\%$ success rate on enough routes averaging 10 steps in length.

D.3. Failure Cases

In this section, we present 2 failure cases in our experiments as mentioned in Sections 5.4 and 6.

Without SFT, RL fails. In Figure 9, we present the training dynamics of failed RL experiments without SFT initialization. We additionally provide output examples of these experiments in Figure 20, where the model tends to generate unstructured response and fail.

RL cannot save overfitted checkpoints. As shown in Figure 19, RL cannot recover the out-of-distribution performance when initialized from a extremely overfitted checkpoint that has an initial per-step accuracy of less than 1%. We additionally provide an output example in Figure 19, where the model fails to adjust to the new rule.

"turn_direction(x)", 其中 $x \in \{\text{'left', 'right', 'slightly left', 'slightly right'}, \text{'forward()'}, \text{'stop()'}\}$

– 最大直行路段长度：任意正整数

动作空间参数适配了 Section 4 中描述的规则变体。对于 Figure 5 所示的实验，我们在训练和域内评估时使用绝对方向动作空间，而在域外评估时使用替代规则。我们设置了最大直行路段长度，以限制转向点之间可移动坐标的数量，防止出现一连串重复的"forward()"动作。我们通过在城市区域训练模型，并在由 Yang et al. (2024a) 发布的基准测试中的全球导航路线上评估其域外性能，进行了视觉分布偏移实验 (Section 5.1)。

Reward design. 当导航智能体在目的地停止或达到最大验证步数 2 时，一个回合终止。奖励函数如下：

- $r = 1$ ：用于在当前坐标生成正确动作
- $r = -1$ ：用于在当前坐标生成错误动作
- $r = -1$ ：用于超过最大验证步数
- $r = -1.5$ ：用于未能检测到地标

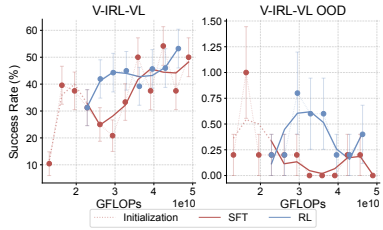


Figure 18: **Overall success rate (%) - GFLOPs for V-IRL-VL under rule variants.** Due to the nature of the task requiring aggregating a trajectory of correct actions, neither training method achieves reasonable out-of-distribution performance.

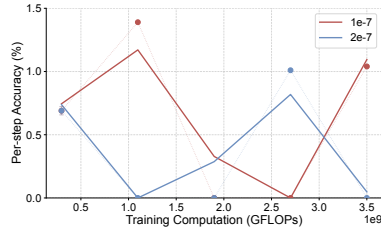


Figure 19: **Out-of-distribution per-step accuracy (%) - GFLOPs for V-IRL-VL under rule variants with overfitted initial checkpoint.** Evaluation metric details can be found in Appendix C.3.

System Prompt (v_0^{in})

[Task Description]

You are an expert 24 points card game player. You are observing these four cards in the image. Note that 'J', 'Q', and 'K' count as '11', '12', and '13' respectively, and each card must be used once. Your goal is to output a formula that evaluates to 24 using numbers from the cards and operators such as '+', '-', '*', '/', '(', ')', and '='.

[Input]

Cards: ['2', 'J', '3', 'K']

[Output]

Your response should be a valid json file in the following format:

```
{
  "cards": [x, y, z, w], where 'J', 'Q', and 'K' count as '10',
  "number": [a, b, c, d], where a, b, c, and d are the numbers on the cards,
  "formula": "an equation that equals 24",
}
```

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

Failed model output (v_t^{out})

To solve this problem, we can use a brute force approach by generating all possible combinations of the cards and checking if any of them can be used to form an equation that equals 24.

Here is a Python solution for this problem:

```
python
def calculate_formula(cards):
    # Define the mapping of cards to their corresponding numbers
    ...
```

Verifier Output (v_t^{ver})

You failed this trial because your formula is incorrect.

$$\triangleright v_{t+1}^{\text{in}} = \text{concat}(v_t^{\text{in}}, v_t^{\text{out}}, v_t^{\text{ver}})$$

Figure 20: **Example model outputs without SFT initialization.** We record model responses using prompts similar to the one shown in Figure 11. The results demonstrate that Llama-3.2-Vision-11B fails to follow instructions properly. We omit the long response which tries to solve the puzzle via code but fails to finish within finite context length.

System Prompt (v_0^{in})

[任务描述]

你是一名导航专家。在观察周围街景的同时，你将收到一系列需要遵循的指令。同时，系统会以文本形式提供你的观察和行动历史记录。你的目标是根据当前观察和指令采取行动。

[指令]

- 首先，向左转面向东方。
- 向前移动，直到到达下一个十字路口，此时 Hotel 32One 应位于你的右后方。
- 向右转面向北方。
- 向前移动，直到到达下一个十字路口，此时 Dragon Gate Chinatown SF 应位于你的右前方。
- 向左转面向东方。
- 向前移动，直到目的地 Café de la Presse 出现在你的右侧。

[当前观察]

你观察到一个包含以下朝向的 2x2 街景图像网格：

[前方，右侧]

[后方，左侧]

你需要判断指令中提到的任何地标是否在该街景网格中可见。

[行动空间]

- "forward()": 表示向前移动 1 步；

- "turn_direction(x)": 表示转向目标朝向，其中 $x \in \text{'north', 'northeast', 'east', 'southeast', 'south', 'southwest', 'west', 'northwest'}$;

- "stop()": 表示导航结束；

[观察与行动序列]

O_1: 附近无地标；

A_1: turn_direction(east)

O_2: 附近无地标；

A_2: forward()

O_3: 附近无地标；

A_3: forward()

O_4: 你观察到包含 4 个视角的图像：你观察到一个十字路口

A_4:

[输出]

你的响应应是一个符合以下格式的有效 json 文件：

```
{
  "current observation": 来自街景网格的最新观察，
  "current instruction": 分析完整指令并确定待执行的句子，
  "action": 从行动空间中选择要采取的行动，
}
```

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

output (v_t^{out})

```
{
  "当前观察": "Hotel 32One 在我右后方；我观察到一个十字路口",
  "当前指令": "右转以面向北方。",
  "动作": "turn_direction(north)",
}
```

Verifier Output (v_t^{ver})

正确解决方案。

$\triangleright$ 正确执行动作后移至下一场景



System Prompt (v_0^{in})**[Task Description]**

You are an expert in navigation. You will receive a sequence of instructions to follow while observing your surrounding street views. You are also provided with your observation and action history in text. Your goal is to first analyze the instruction and identify the next sentence to be executed. Then, you need to provide the action to be taken based on the current observation and instruction.

[Instruction]

1. First, turn right to face northwest.
2. Move forward until you reach next intersection where Korean War Memorial is on your left.
3. Turn left to face southwest.
4. Move forward until you reach next intersection where Korean War Memorial is on your left behind.
5. Turn right to face north.
6. Move forward until you reach next intersection.
7. Turn left to face east.
8. Move forward until you reach next intersection.
9. Turn left to face north.
10. Move forward until you reach next intersection.
11. Turn right to face east.
12. Move forward until you reach next intersection.
13. Turn left to face northeast.
14. Move forward until you reach next intersection where 9/11 Memorial & Museum is on your left.
15. Turn right to face northwest.
16. Move forward until you reach destination where The destination 9/11 Memorial & Museum is on your right front.

[Action space]

"forward()": indicates moving forward one step

"turn_direction(x)": indicates adjust the ego agent direction towards x direction. x could be any following ['left', 'right', 'slightly left', 'slightly right']

"stop()": indicates the navigation is finished.

[Observations and actions sequence]

O_1: You observe an image of 4 views;

A_1:

[Output]

```
{
  "current observation": latest observation from the streetview grid,
  "current instruction": analyze the full instruction and identify the sentence to be executed,
  "action": the action to be taken chosen from the action space,
}
```

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

Failed model output (v_t^{out})

```
{
  "current observation": "No landmarks nearby",
  "current instruction": "Turn right to face northwest.",
  "action": "turn_direction(northwest)",
}
```

Verifier Output (v_t^{ver})

Incorrect action.

▷ Fail this sample.



Figure 21: **Failed example of V-IRL transition due to overfitting.** This phenomenon happens more frequently during scaling up supervised fine-tuning.

System Prompt (v_0^{in})**[任务描述]**

你是一名导航专家。你将收到一系列需要遵循的指令。同时，系统会以文本形式提供你的观察和行动历史。你的目标是首先分析指令，并确定接下来要执行的句子。然后，你需要根据当前观察和指令提供要采取的动作。

[指令]

1. 首先，左转以面向东方。
2. 向前移动，直到到达下一个十字路口，此时 Hotel 32One 在你的右后方。
3. 右转以面向北方。
4. 向前移动，直到到达下一个十字路口，此时 Dragon Gate Chinatown SF 在你的右前方。
5. 左转以面向东方。
6. 向前移动，直到目的地 Café de la Presse 在你的右侧。

[动作空间]

- "forward()": 表示向前移动 1 步；

- "turn_direction(x)": 表示转向目标航向，其中 $x \in \{\text{'north', 'northeast', 'east', 'southeast', 'south', 'southwest', 'west', 'northwest'}\}$;

- "stop()": 表示导航完成；

[观察与动作序列]

O_1: 附近无地标；

A_1: turn_direction(east)

O_2: 附近无地标；

A_2: forward()

O_3: 附近无地标；

A_3: forward()

O_4: Hotel 32One 在你的右后方；你观察到一个十字路口

A_4:

[输出]

你的响应是一个有效的 JSON 文件，格式如下：

```
{
  "当前观察": 来自街景网格的最新观察，
  "当前指令": 分析完整指令并确定要执行的句子，
  "动作": 从动作空间中选择要执行的动作，
}
```

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

output (v_t^{out})

```
{
  "当前观察": "Hotel 32One 在我右后方；我观察到一个十字路口",
  "当前指令": "右转向北方。",
  "动作": "turn_direction(north)",
}
```

Verifier Output (v_t^{ver})

正确解决方案。

▷ 正确动作后移至下一场景

Figure 14: 我们提示更新的一个示例，用于使用 v_{t+1}^{in} 和 $v_t^{\text{in}}, v_t^{\text{out}}$ 构建 v_t^{ver} 。棕色部分标记了任务及相关信息，紫色部分表示状态 (s_t) 特定信息。其中棕色部分标记任务及相关信息，紫色部分表示状态 (s_t) 特定信息。蓝色和红色分别描述来自模型和验证器的输出。

C. Experimental Setup

本节详细说明了 Section 5 中使用的实验设置。我们首先描述用于监督微调 (Appendix C.1) 的数据收集设置。然后，我们介绍训练流程 (Appendix C.2)。最后，我们描述了评估指标以及用于生成图表的统计工具 (Appendix C.3)。

C.1. Data

SFT data collection. 如 Figures 11 to 14 所示, GeneralPoints 和 V-IRL 环境天然与提示-响应对话结构对齐。我们通过将每个系统提示与其对应的专家响应配对来创建训练样本。正文中的所有 SFT 实验均使用最优的单轮提示-响应对, 不包含任何验证或修订步骤。

SFT on sub-optimal trajectories 为了探究更多样化的 SFT 数据如何影响 SFT 的分布外性能, 我们在 GP-L 上使用次优轨迹作为训练数据进行了一项消融研究。与专家提示-响应对不同, 这些次优轨迹在其提示中包含错误和验证信息。这种格式与允许进行多次验证迭代的评估场景相一致, 类似于用于下游 RL 训练的数据。在 Figure 15 中, 我们观察到 SFT 仍然只是记忆训练数据, 且分布外性能下降。这一证据表明, 记忆的发生是由于 SFT 训练的根本性质, 而非 SFT 数据本身。

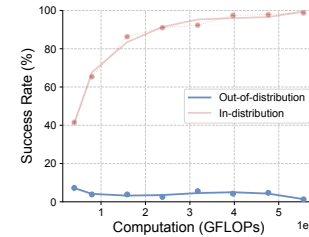


Figure 15: SFT experiments on GP-L using sub-optimal trajectories. 与 Figure 5 中的结果类似, 即使我们增加了轨迹的多样性, SFT 仍然过拟合训练数据。

C.2. Training Pipeline

如 Section 5 所示, 我们遵循 RL4VLM (Zhai et al., 2024a) 的训练流程, 首先用 SFT 初始化模型, 然后分别扩大 SFT 和 RL (Schulman et al., 2017) 的计算规模, 均从此初始化模型开始。对于正文中所有的 SFT 和 RL 实验, 我们为每个实验使用统一的学习率来调整所有组件。所有训练实验均在配备 8 块 H800 (80GB) 的机器上进行。

C.3. Evaluation Metric

Per-step accuracy. 我们在 V-IRL-VL 任务中报告了每步准确率, 如 Figures 5 and 6 所示。当模型选择的动作与该位置上的专家轨迹匹配时, 单个步骤被视为正确。请注意, 中间验证步骤在此处被计为独立样本。

Success rate. 我们在 Figures 5 and 6 中报告了 GP-L、GP-VL、V-IRL-L 和 V-IRL-VL 的成功率 (%)。在 GeneralPoints 任务中, 成功被定义为在推理时间验证期间至少成功一次。在 V-IRL 任务中, 当模型在路径上的每个可移动点都采取正确动作时, 样本被记录为成功。

Computation estimation. 我们按照与 (Snell et al., 2024; Hoffmann et al., 2023) 类似的方式估算了训练 X 的 FLOPs, 其中 $X_{train} = 6ND_{train}$ 且 $X_{inference} = 2ND_{inference}$ 。此处, N 代表模型参数, D_{train} 代表训练期间的令牌数量。假设我们的 SFT 和 RL 实验从一个在 D_{init} 个令牌上训练得到的检查点开始, 我们可以通过以下方程估算 SFT 和 RL 的训练计算量:

$$X_{SFT} = 6N(D_{init} + D_{SFT})$$

$$X_{RL} = 6N(D_{init} + D_{RL}) + 2ND_{buffer}$$

请注意, 所使用的在线策略 RL 算法 PPO (Schulman et al., 2017) 包含回放缓冲区收集和优化的迭代阶段, 因此需要额外的推理计算。为简化起见, 我们通过

以下方式近似该计算项：

$$\begin{aligned} D_{buffer} &\approx \frac{E\bar{d}_i\bar{d}_o}{D_{RL}} \cdot D_{RL} \\ &= \lambda D_{RL} \end{aligned}$$

其中 $E \in \mathbb{N}$ 表示自回归生成过程的数量， $\bar{d}_i, \bar{d}_o$ 表示平均输入令牌和输出令牌。经计算，我们估计 GeneralPoints 和 V-IRL 的 λ 分别为 6 和 5.1。

Line smoothing and error bar. 我们论文中的所有折线图均采用多项式阶数为 3 的 Savitzky-Golay 滤波器作为平滑函数。我们假设每个评估数据点服从二项分布，并使用 $\sqrt{\frac{P(1-P)}{N}}$ 近似标准误差，其中 P 是十进制成功率， N 是样本数量。

D. Additional Experimental Results

在本节中，我们提供正文中未涵盖的额外实验结果。

D.1. Ablation Studies on GP-VL

如 Section 6 所述，我们观察到一个异常现象：SFT 未能达到与 RL 相当的内分布性能（参见 Figure 5 子图第 1 行第 3 列）。为了进一步探究此问题，我们对不同的超参数选择进行了消融研究。

SFT. 我们在 Section 5 中 GP-VL 的相同任务设置下，对超参数选择进行了消融。对于微调所有参数的实验，我们搜索了学习率范围 $\{1 \times 10^{-4}, 1 \times 10^{-4}, 1 \times 10^{-5}, 1 \times 10^{-6}, 5 \times 10^{-7}, 1 \times 10^{-7}\}$ 。冻结视觉编码器时，我们搜索学习率 $\{1 \times 10^{-6}, 1 \times 10^{-7}\}$ 。冻结视觉编码器和适配器时，我们搜索学习率 $\{1 \times 10^{-6}, 5 \times 10^{-7}, 1 \times 10^{-7}\}$ 。我们在 Figure 16 中提供了内分布成功率曲线。

RL. 为 RL 实验寻找合适的超参数只需付出最小的努力。我们搜索了学习率 $2 \times 10^{-6}, 1 \times 10^{-6}$ ，其内分布成功率曲线如 Figure 17 所示。在我们的 RL 实验中，所有参数都是可调的。

D.2. More results on V-IRL-VL

呼应 Figure 5 中的每步准确率结果，我们在 Figure 18 中报告了 V-IRL-VL 的总体成功率。由于任务的复杂性，两种训练方法实现的总体成功率均未超过 1%。对于 V-IRL 而言，总体成功率是一个要求高得多的指标，因为它聚合了每步误差。例如，一个每步准确率为 10% 的随机策略，在平均长度为 10 步的足够多路线上，其成功率大约仅为 $10^{-8}\%$ 。在本节中，我们将展示实验中遇到的 2 个失败案例，如 Sections 5.3 and 6 所述。

Without SFT, RL fails. 在 Figure 9 中，我们展示了未经 SFT 初始化的失败 RL 实验的训练动态。我们还在 Figure 20 中提供了这些实验的输出示例，其中模型倾向于生成非结构化的响应并失败。

RL cannot save overfitted checkpoints. 如 Figure 19 所示，当从一个初始单步准确率低于 1% 的极度过拟合检查点初始化时，RL 无法恢复其分布外性能。我们还在 Figure 19 中提供了一个输出示例，其中模型未能适应新规则。

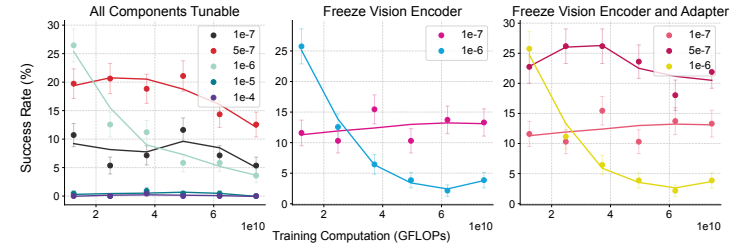


Figure 16: Ablation studies on **GeneralPoints-VL SFT**. 我们消融了学习率，并报告了所有实验的分布内情节成功率 (%)。所有实验均未显示出超过 30% 成功率的增长趋势。

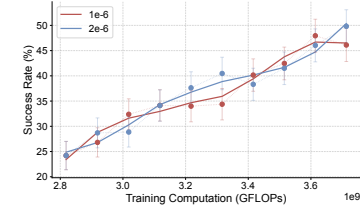


Figure 17: Ablation studies on **GeneralPoints-VL RL**. 呼应 Figure 16，我们消融了学习率并报告了两个实验的内分布回合成功率 (%)。此处所有组件都是可调的。

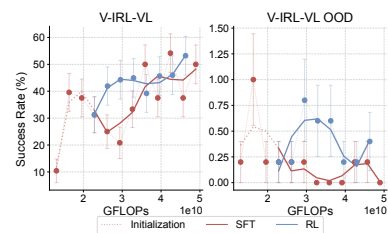


Figure 18: **Overall success rate (%) - GFLOPs for V-IRL-VL** 在不同规则变体下的表现。由于任务本质要求聚合一系列正确动作，两种训练方法均未获得合理的分布外性能。

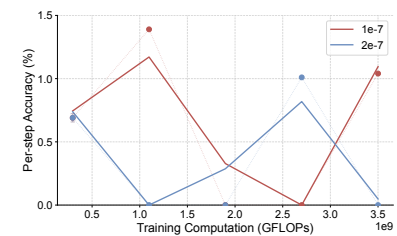


Figure 19: **Out-of-distribution per-step accuracy (%) - GFLOPs for V-IRL-VL** 在使用过拟合初始检查点的规则变体下的表现。评估指标详情可在 [Appendix C.3](#) 中找到。

System Prompt (v_0^{in})

[Task Description]

You are an expert 24 points card game player. You are observing these four cards in the image. Note that 'J', 'Q', and 'K' count as '11', '12', and '13' respectively, and each card must be used once. Your goal is to output a formula that evaluates to 24 using numbers from the cards and operators such as '+', '-', '*', '/', '(', ')', and '='.

[Input]

Cards: ['2', 'J', '3', 'K']

[Output]

Your response should be a valid json file in the following format:

```
{
  "cards": [x, y, z, w], where 'J', 'Q', and 'K' count as '10',
  "number": [a, b, c, d], where a, b, c, and d are the numbers on the cards,
  "formula": "an equation that equals 24",
}
```

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \quad \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

Failed model output (v_t^{out})

To solve this problem, we can use a brute force approach by generating all possible combinations of the cards and checking if any of them can be used to form an equation that equals 24.

Here is a Python solution for this problem:

```
""python
def calculate_formula(cards):
# Define the mapping of cards to their corresponding numbers
...
```

Verifier Output (v_t^{ver})

You failed this trial because your formula is incorrect.

$$\triangleright v_{t+1}^{\text{in}} = \text{concat}(v_t^{\text{in}}, v_t^{\text{out}}, v_t^{\text{ver}})$$

Figure 20: **Example model outputs without SFT initialization.** 我们使用与 Figure 11 所示类似的提示词记录模型响应。结果表明，Llama-3.2-Vision-11B 未能正确遵循指令。我们省略了试图通过代码解决谜题但因上下文长度有限而未能完成的长篇响应。

System Prompt (v_0^{in})

[任务描述]

你是一名导航专家。在观察周围街景的同时，你将收到一系列需要遵循的指令。同时，系统会以文本形式提供你的观察和行动历史。你的目标是首先分析指令，识别出接下来要执行的句子。然后，根据当前观察和指令，提供要采取的行动。

[指令]

1. 首先，向右转面向西北方。
2. 向前移动，直到到达下一个十字路口，此时朝鲜战争纪念碑在你的左侧。
3. 向左转面向西南方。
4. 向前移动，直到到达下一个十字路口，此时朝鲜战争纪念碑在你的左后方。
5. 向右转面向北方。
6. 向前移动，直到到达下一个十字路口。
7. 向左转面向东方。
8. 向前移动，直到到达下一个十字路口。
9. 向左转面向北方。
10. 向前移动，直到到达下一个十字路口。
11. 向右转面向东方。
12. 向前移动，直到到达下一个十字路口。
13. 向左转面向东北方。
14. 向前移动，直到到达下一个十字路口，此时 9/11 纪念 & 博物馆在你的左侧。
15. 向右转面向西北方。
16. 向前移动，直到到达目的地，此时目标地点 9/11 纪念 & 博物馆在你的右前方。

[行动空间]

"forward()": 表示向前移动一步

"turn_direction(x)": 表示将智能体自身方向调整为朝向 x 方向。x 可以是以下任意一项 ['left', 'right', 'slightly left', 'slightly right']

"stop()": 表示导航完成。

[观察与行动序列]

O_1: 你观察到一张包含 4 个视角的图像：

A_1:

[输出]

{

"current observation": 来自街景网格的最新观察，

"current instruction": 分析完整指令并识别出要执行的句子，

"action": 从行动空间中选择要采取的行动，

}

Appending model and verifier outputs to obtain v_t^{in}

$$v_t^{\text{in}} = [v_0^{\text{out}}, v_0^{\text{ver}}, v_1^{\text{out}}, v_1^{\text{ver}}, \dots, v_{t-1}^{\text{out}}, v_{t-1}^{\text{ver}}] \triangleright v_t^{\text{in}} = \text{concat}(v_0^{\text{in}}, [v_k^{\text{out}}, v_k^{\text{ver}}]_{k=0}^{t-1})$$

model output (v_t^{out})

{

"当前观察": "附近没有地标",

"当前指令": "向右转面向西北方向。",

"动作": "turn_direction(northwest)",

}

Verifier Output (v_t^{ver})

动作错误。

▷ 此样本失败。



Failed

Figure 21: **Failed example of V-IRL** 由于过拟合导致的转变。这种现象在扩大监督微调规模时更频繁地发生。